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# The Impact of Levothyroxine in Women with Positive Thyroid Antibodies on Pregnancy Outcomes: A Systematic Review and Meta-Analysis of Randomised Controlled Trials



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# The Impact of Levothyroxine in Women with Positive Thyroid Antibodies on Pregnancy Outcomes: A Systematic Review and Meta-Analysis of Randomised Controlled Trials

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## ABSTRACT

**Objective:** To evaluate the effect of levothyroxine therapy on pregnancy outcomes compared with placebo or no treatment in women without overt hypothyroidism who are TPOAb and/or TgAb-positive.

**Design:** Systematic review and meta-analysis of randomised controlled trials

**Study Eligibility Criteria**: Pre-specified criteria for inclusion were: randomised trials of levothyroxine versus control (placebo or no treatment) among women with positive TPOAb or TgAb who were pregnant or considering conception.

**Data Sources**: Ovid MEDLINE, EMBASE, CINAHL, Cochrane Database of Systematic Reviews, and Cochrane Central Register of Controlled Trials were searched from 1980 to April 17, 2019.

**Outcome Measures**: Pre-specified data elements were extracted and where appropriate, meta-analyses were conducted. Main outcomes include pregnancy achieved, miscarriage, preterm delivery and live birth.

**Risk of bias assessment:** Cochrane Risk of Bias Tool (Modified) for Quality Assessment of Randomised Controlled Trials.

**Results:** From 2,812 citations, 79 citations were identified for full text review. Of these, six trials (total of 2,263 women) were included for qualitative and quantitative analyses. Risk of bias was deemed low for only one trial. There was no significant difference in the relative risk (RR) of pregnancy achieved (RR 1.03; 95% CI 0.93, 1.13), miscarriage (RR 0.93; 95% CI 0.76, 1.14), preterm delivery (RR 0.66; 95% CI 0.39, 1.10), or live births (RR 1.01; 95% CI 0.89, 1.16) in thyroid autoimmune women treated with levothyroxine compared to controls. Sensitivity analyses of preterm birth identified study quality and timing of levothyroxine initiation as sources of heterogeneity. There was evidence of publication bias.

**Conclusions:** Among pregnant women or women planning conception, with thyroid autoimmunity, there is a lack of evidence of benefit for levothyroxine use. Recommendations to use levothyroxine in this setting need to be reconsidered.


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**Strengths and limitations of this study**

- This systematic review summarises the most current published randomised controlled trials not included in other systematic reviews and meta-analysis.
- To minimize the bias associated with observational studies, only randomised controlled trials were included.
- This systematic review and meta-analysis is rigorous in design, methodology and analyses with a prepublished protocol.
- Of the six trials identified for inclusion, only one was deemed low risk of bias.

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**Competing interests statement:** The authors have no competing interests to declare.

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**Contribution to authorship:** PL and LED conceived the idea for this systematic review and meta-analysis, and conducted the literature search along with the initial title/abstract review. JMY contributed her expertise in systematic review methodology to the design and conduct of this systematic review. LWL and JLB conducted the full text review, study selection, data extraction, and statistical analyses. LWL wrote the first draft of the manuscript. All authors contributed to critical review and approval of the final manuscript.

**Ethics Statement.** The authors adhered to the Preferred Reporting Items for Systemic Reviews and Meta-Analyses (PRISMA) guidelines. Ethical committee approval was not required for this study.

**PROSPERO registration number** CRD42019130459

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## INTRODUCTION:

Thyroid autoimmunity, defined by the presence of thyroid peroxidase antibodies (TPOAb) and/or thyroglobulin antibodies (TgAb), is relatively common in pregnant women. The prevalence of TPOAb positivity in the first trimester ranges from 5 to 15% in unselected pregnant women[1 2]. The prevalence of thyroid autoimmunity is even higher in women with a history of recurrent pregnancy loss, reported as 23 to 31%[3 4].

The association between thyroid autoimmunity and poor pregnancy outcomes, specifically miscarriage and preterm birth, has been well described[5 6]. However, the mechanism linking thyroid autoimmunity with increased risk of miscarriage and preterm birth in women without overt hypothyroidism is unclear. Two competing mechanistic theories propose that 1) thyroid autoantibodies are a marker of a hostile immune environment, or 2) thyroid autoantibodies act as a surrogate biomarker of impaired thyroid hormone reserves, despite normal thyroid function tests[7]. If the latter theory is true, one could postulate that supplementation with levothyroxine may improve thyroid hormone levels and prevent negative pregnancy outcomes.

Whether or not levothyroxine therapy, either prior to or during pregnancy, improves pregnancy outcomes in women with thyroid autoimmunity is controversial. The 2017 American Thyroid Association (ATA) pregnancy guidelines strongly recommends levothyroxine for women with thyroid autoimmunity and a thyroid stimulated hormone (TSH) above the pregnancy-specific reference range. These guidelines recommend a consideration of levothyroxine in thyroid autoimmune women with TSH above 2.5 mIU/L and among euthyroid women with positive TPOAb and a history of pregnancy loss[8]. The evidence evaluating levothyroxine treatment in

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pregnant women with thyroid autoantibodies is conflicting. Additionally, the limitations of observational studies must be considered. Observational studies identify associations, but do not provide the highest level of evidence for determining causation or therapeutic efficacy. Randomised controlled trials are considered the gold standard to evaluate therapeutic efficacy[9]. Therefore, we performed a systematic review and meta-analysis to evaluate all randomised controlled trials of levothyroxine therapy in women with thyroid autoimmunity who were pregnant or planning pregnancy.

## METHODS

### Protocol and Registration

A systematic review and meta-analysis was performed as outlined in the registered protocol (PROSPERO CRD42019130459), and is reported in accordance with the Preferred Reporting Items for Reviews and Meta-Analyses (PRISMA).

### Data Sources and Search Strategy

The following databases were searched in duplicate on April 17, 2019 from 1980: Ovid MEDLINE, EMBASE, CINAHL, Cochrane Database of Systematic Reviews, and Cochrane Central Register of Controlled Trials. Clinicaltrials.gov was searched for active trials. The search strategy was developed in consultation with a medical librarian and used a combination of keywords related to hypothyroidism or thyroid autoimmunity and pregnancy (Supplemental S1). The search was conducted using an adapted version of the Cochrane search filter for randomised controlled trials where possible and was limited to human studies[10].

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**Study Selection**

Studies that met all the following criteria were included in this review: (1) pregnant women or women contemplating pregnancy (including patients seeking assisted reproductive technologies), with positive TPOAb or TgAb, euthyroid or with subclinical hypothyroidism (by any prespecified study definition); (2) randomised to levothyroxine versus control (placebo or no treatment); (3) reported pregnancy outcomes; (4) reported in either English or French; (5) published from 1980 to April 17, 2019. The publication year 1980 was specifically chosen to coincide with the availability of more sensitive TSH assays[11 12].

Study selection was completed in two separate stages by two independent reviewers (PL and LED or LWL and JLB). First, duplicates were removed, and titles and abstracts were screened for eligibility (by PL and LED). Second, full citations deemed eligible at the title and abstract stage were retrieved and reviewed for eligibility in a full-text format (by LWL and JLB). Inclusion criteria was applied to select eligible articles and reasons for exclusion were recorded. Agreement was recorded at each stage and reported as a kappa statistic[13]. All disagreements between reviewers were solved by consensus, and, if needed, by discussion with a third independent reviewer (LED).

**Data Collection Process**

Relevant study information was extracted by two independent reviewers (LWL and JLB) using standardized data extraction forms in Microsoft Excel (Version 16.16.18, Microsoft Corporation, Redmon, WA, USA). Extracted data elements included: study design, country, participant inclusion and exclusion criteria, TPOAb and TgAb results, definition of TPOAb and TgAb

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positivity, TPOAb and TgAb assays used, thyroid stimulating hormone (TSH) and thyroxine (T4) assays used and normal ranges, baseline serum concentrations of TSH and T4, iodine status of the country, levothyroxine dosing, achieved TSH and T4 after treatment, gestational timing of levothyroxine initiation, duration of treatment, details of the intervention, quality of the methods, inclusion of patients with subclinical hypothyroidism and definition of subclinical hypothyroidism, patient baseline characteristics, number of patients included in each group, pregnancy outcomes and their definitions.

**Quality Assessment**

Cochrane Risk of Bias Tool (Modified) for Quality Assessment of Randomised Controlled Trials[14] was used to assess study quality and evaluate the risk of bias by two independent reviewers (LL and JB). This tool assesses seven methodologic domains and these components enable classification of randomised trials as "Low Risk of Bias", "High Risk of Bias" or "Unclear Risk of Bias".

**Study Outcomes**

The following pregnancy outcomes were prespecified and included: pregnancy by trial definition, miscarriage, preterm delivery, live-birth, birth weight, large for gestational age, small for gestational age, gestational age at delivery, multiple pregnancy, stillbirth, neonatal intensive care unit admission, preeclampsia, and gestational diabetes.

**Data Synthesis and Analysis**

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Where appropriate, meta-analyses were conducted to evaluate the relative risk of pre-specified pregnancy outcomes including pregnancy achieved, miscarriage, preterm delivery and live-birth in participants randomised to levothyroxine therapy compared with placebo or no treatment. Random-effect models were used. Relative risk (RR) and 95% confidence intervals (95% CI) were calculated. Statistical heterogeneity was quantified using $I^2$ statistics, where $I^2$ >50% represents moderate and $I^2$ >75% represents substantial heterogeneity across studies. The following prespecified sensitivity analyses were conducted to explore heterogeneity: 1) TSH levels at baseline, 2) time of levothyroxine initiation (preconception or during pregnancy), 3) use and type of assisted reproduction technologies, 4) maternal characteristics, and 5) study quality. Publication bias was assessed using graphical aids. Statistical analysis was performed using STATA version 14 (StataCorp, College Station, TX, USA) and Review Manager 5.3 (Version 5.3.5, The Cochrane Collaboration, Copenhagen, Denmark).

**Patient and Public Involvement**

No patient involved.

**RESULTS**

The search results are summarised in Figure 1. Among the 2,812 citations reviewed for title and abstract, 79 citations were identified for full text review. Six randomised controlled trials (total of 2,263 women) met all eligibility criteria and were included in the qualitative and quantitative analyses[15-20]. There was a substantial level of agreement for both the title and abstract review ($\kappa$=0.64) and for the full text review ($\kappa$=0.70).

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## Trial Characteristics

Trial characteristics are summarised in Table 1. Three trials initiated levothyroxine therapy prior to conception[15 18 20], whereas three trials initiated therapy in the first trimester[16 17 19]. Two trials compared levothyroxine to placebo[15 17], and the other four trials compared with no treatment[16 18-20]. Reproductive technologies were used in three trials[15 18 20], with one trial not specifying details with this regard. All trials included women with positive TPOAb with only one trial including TPO or Tg antibodies[20]. The TSH normal reference ranges and definition of TPO positivity were prespecified in each trial and are summarised in Table 1.

## Patient Characteristics

Baseline participant characteristics are summarised in Table 1 and Supplemental S2. Mean maternal age ranged from 26 years[16] to 32 years[15]. Mean reported body mass index ranged from 22.7 kg/m$^2$ [20] to 26.5 kg/m$^2$ [15]. Baseline TSH values varied according to assay specific reference ranges. Baseline free T4 values were comparable in the three trials that reported this information[15 18 19] (Supplemental S2).

## Risk of Bias Assessment

Risk of bias assessment is presented in Figure 2. Only one trial[15] was considered "low risk" of bias in all seven domains. Four trials[15 17-19] were assessed at "low risk" for selection bias. Two trials[16 20] were "unclear risk of bias" in allocation concealment. In four trials comparing levothyroxine to no treatment[16 17 19 20], the assessors were not blinded to patient randomization and thus these trials were considered to be "high risk" for performance bias. Published trial

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| | | | | | | | | | Mean GA of LT4 initiation (weeks) | | Baseline TSH (mIU/L) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Study Author (year)/ Country** | **N** | **Intervention levothyroxine** | **Control** | **Use of assisted reproduction technologies (% of participants)** | **Inclusion of subclinical hypothyroidism** | **TSH (Normal Range; mIU/L)** | **Defintion of TPOAb positivity (IU/mL)** | | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm |
| Dhillon-Smith *et al.* (2019)/ United Kingdom | 952 | Initiated preconception | Placebo | Yes (45%) | No | 0.44-3.63 | > 99% concordance with UK NEQAS IIA | | Preconception | Preconception | 2.1 | 2 |
| Nazarpour *et al.* (2017)/ Iran | 131 | Initiated 4-8 days following first prenatal visit | No treatment | Unclear | Yes | 0.1-2.5 | >50 | | 11.4 ± 4.2 | 11.4 ± 4.2 | 3.7 | 3.2 |
| Negro *et al.* (2005)/ Italy | 72 | Initiated 1 month before assisted reproduction technologies | No treatment | Yes (100%) | Yes | 0.27-3.2 | >100 | | Preconception | Preconception | 1.9 | 1.7 |
| Negro *et al.* (2006)/ Italy | 115 | Initiated at first endocrinological visit (2-7d after first obstetrics visit) | Placebo | No | No | 0.27-4.2 | >100 | | 10.4 ± 3.1 | 10.3 ± 3.1 | 1.6 | 1.7 |
| Negro *et al.* (2016)/ Italy | 393 | Initiated in first trimester prior to 12 weeks | No treatment | No | No | 0.5-2.5 | >16 | | 7.1 ± 1.2 | 7.0 ± 1.2 | 1.4 | 1.4 |
| Wang *et al.* (2017)/ China | 600 | Initiated 2-4 weeks prior to controlled ovarian hyperstimulation | No treatment | Yes (100%) | No | 0.45-4.78 | 60 | | Preconception | Preconception | 2.9 | 2.1 |

**Table 1.** Characteristics of included randomised controlled trials. N, sample size; TSH, thyroid stimulating hormone; TPOAb, thyroid peroxidase antibody; LT4, levothyroxine; GA, gestational age; FT4, free thyroxine

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protocols were located for only two trials[15 20]. Thus, only these two trials were assessed as "low risk" for reporting bias.

**Pregnancy Outcomes**

There were sufficient data for meta-analyses of four pregnancy outcomes, as summarised in Figure 3. These outcomes include pregnancy achieved, miscarriage, preterm delivery and live births.

Pregnancy achieved was reported in three trials[15 18 20] (n=1,626) that initiated levothyroxine pre-conception. There was no significant difference in pregnancy achieved in thyroid autoimmune women treated levothyroxine compared with control (RR 1.03 [95% CI 0.93, 1.13]; $I^2$=0%)

Miscarriage was reported in all six trials (n=1,427) and was defined as pregnancy loss at less than 20 weeks[16], less than 24 weeks[15], less than 28wks (early miscarriage defined as first 12 weeks, late miscarriage defined as between 13-28 weeks)[20], or not clearly defined[17-19]. There was no significant difference in the relative risk (RR) of miscarriage in TPOAb-positive pregnant women treated with levothyroxine compared with control (RR 0.93 [95% CI 0.76, 1.14]; $I^2$=0%). Inclusion of trials[15 18 20] that initiated levothyroxine only preconception revealed no significant difference compared with control (RR 0.91 [95% CI 0.72, 1.15]; $I^2$=0%).

Preterm delivery was reported in five trials (n= 1,354)[15-17 19 20]. Preterm delivery was defined as live birth prior to 34 weeks[15] or 37 weeks[16 17 19 20]. There was no significant difference in the

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relative risk of preterm delivery in thyroid autoimmune pregnant women treated with levothyroxine compared with control (RR 0.66 [95% CI 0.39, 1.10]; $I^2$=53%).

Live births were reported in two trials (n= 1,540)[15 20]. Live birth was defined as delivery of a living fetus beyond 27 weeks[20] or beyond 34 weeks[15]. There was no significant difference in live births in thyroid autoimmune women treated with levothyroxine compared with control (RR 1.01 [95% CI 0.89, 1.16]; $I^2$ = 0%).

Only one trial[17] reported on pre-eclampsia and found no difference in this outcome among thyroid autoimmune women treated with levothyroxine compared with control. None of the studies reported the outcome of gestational diabetes.

**Neonatal Outcomes:**

Neonatal outcomes were only reported in two trials (n = 1,083)[15 16]. Meta-analysis for gestational age at time of delivery (measured in days) and birth weight (measured in grams) are summarised in Supplemental S3. The weighted mean difference for gestational age at time of delivery in women treated with levothyroxine compared with placebo or no treatment was 2.35 days (95% CI -4.51, 9.20; $I^2$ = 90%). Weighted mean difference for birth weight in women treated with levothyroxine compared with placebo or no treatment was -24.05g (95% CI -96.91, 48.80; $I^2$ = 0%).

Only one trial[16] reported neonatal intensive care unit admission. They found fewer admissions among thyroid autoimmune women treated with levothyroxine (2 of 56 compared with control

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12 of 58, $p < 0.05$). No studies reported large for gestational age, small for gestational age or multiple pregnancy.

**Sensitivity Analyses**

A sensitivity analysis was performed to explore the heterogeneity of our meta-analysis for preterm delivery. Results are summarised in Supplemental S4.

TSH levels at baseline were subdivided into TSH ≤ 2.5 mIU/mL[15 17 19] and TSH > 2.5 mIU/mL[16 20]. The relative risk of preterm delivery in pregnant women with baseline TSH ≤ 2.5 mIU/mL was 0.64 (95% CI 0.36, 1.14; $I^2$=32%) compared with a relative risk of 0.62 (95% CI 0.17, 2.33; $I^2$=80%) in pregnant women with baseline TSH > 2.5 mIU/mL.

Timing of levothyroxine initiation was stratified into preconception[15 20] and in pregnancy[16 17 19]. The relative risk of preterm delivery in women initiated on levothyroxine preconception was 1.10 (95% CI 0.69, 1.75; $I^2$=0%) compared a relative risk of 0.46 (95% CI 0.28, 0.75; $I^2$=2%) in women initiated on levothyroxine in the first trimester.

The relative risk of preterm delivery in women using assisted reproductive technologies (ART)[15 20] was 1.10 (95% CI 0.69, 1.75, $I^2$ = 0%) compared with a relative risk of 0.51 (95% CI 0.26, 1.02; $I^2$ = 27%) in women that did not use ART[17 19].

Sensitivity analyses was also performed to explore the impact of fixed[15 18] and adjusted levothyroxine[16 17 19 20] doses on miscarriage rates (Supplemental S5). The relative risk of fixed

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levothyroxine dose was 0.89 (95% CI 0.66, 1.20; $I^2$=12%) compared with adjusted levothyroxine dose of 0.95 (95% CI 0.68, 1.33; $I^2$ = 0%).

Study quality was defined as high if there was low risk of bias across all seven domains of the Cochrane Collaboration's tool. We were unable to perform a sensitivity analysis exploring the impact of risk of bias on preterm delivery, as only one trial met the definition for high quality per the Cochrane Collaboration tool[15]. A funnel plot for preterm delivery is shown in Supplemental S6. On visual inspection, there was asymmetry indicating potential publication bias.

## DISCUSSION

This systematic review and meta-analysis of randomised controlled trials of levothyroxine therapy in women with thyroid autoimmunity without overt thyroid dysfunction demonstrates no evidence of benefit of levothyroxine therapy across prespecified pregnancy outcomes. Specifically, treatment with levothyroxine did not improve pregnancies achieved, miscarriage preterm delivery or live-birth across a total of 2,263 women, inclusive of those with a history of pregnancy loss and use of reproductive technologies. Furthermore, the availability of larger and better-quality trials since the publications of the American Thyroid Association 2017 pregnancy recommendations signals the need to update these guidelines as they pertain to the treatment of thyroid autoimmune women.

Small randomised clinical trials initially demonstrated that levothyroxine therapy reduced miscarriage and/or pre-term delivery in TPOAb-positive women compared with placebo or no

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treatment intervention[16-18]. However, the unclear definition of miscarriage and pre-term delivery among these early trials means that selective reporting of outcomes could have biased their reported findings. Subsequent large, multi-center randomised controlled trials with clear prespecified outcome definitions have shown no impact of levothyroxine therapy on maternal or fetal outcomes compared with no levothyroxine treatment in pregnant women with subclinical hypothyroidism, isolated hypothyroxinemia or positive TPOAb[15 20-22]. These intervention trials involved pregnant women or women seeking pregnancy, including individuals with a history of pregnancy loss.

A systematic review and meta-analysis of a large study population (n = 47,045) of individual participant and population data demonstrated increased risk of preterm birth in TPOAb-positive women[6]. In light of these findings, Cappola and Casey[23] highlighted the need to address whether levothyroxine treatment decreases the risk of preterm birth among women with TPOAb. Three other systematic reviews and meta-analyses examined the effect of levothyroxine supplementation in pregnant women with thyroid autoimmunity[24-26]. Unlike our review which only included randomised controlled trials, two of the other reviews[24,25] included observational studies. The findings of our meta-analysis were consistent with the review by Sun *et al.*[24], which showed no benefit of levothyroxine supplementation on their pregnancy outcomes. In contrast, Rao *et al.*[25 26] demonstrated a beneficial effect of levothyroxine treatment on reducing pregnancy loss and preterm birth among pregnant women with subclinical hypothyroidism and/or thyroid autoimmunity. Several notable distinctions likely account for these discrepant results. First, the Dhillon-Smith *et al.*[15] publication was not included is these reviews. Furthermore, both reviews by Rao *et al.*[25 26] included a trial that strongly influenced their results but did not report outcome

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data in sufficient detail to be included in our systematic review[27]. Our attempts to clarify the outcome data of this trial(27) with the authors were left unresponded.

Most miscarriages occur early in pregnancy, prior to the first obstetrics appointment, thus initiation of levothyroxine prior to conception best addresses its impact on miscarriage rates. The pathophysiology underlying the association between thyroid autoimmunity in euthyroid pregnant women and miscarriage is still not completely understood. One hypothesis postulates that thyroid autoimmunity reflects inadequate thyroid reserves to accommodate for increased physiologic demands of pregnancy[28]. It has been postulated that delayed initiation of levothyroxine in pregnancy may account for its failure to reduce miscarriage. Our meta-analysis found that initiation of levothyroxine preconception or in the first trimester did not impact miscarriage outcomes. This suggests that inadequate thyroid reserve in women with thyroid autoimmunity does not account for the increased risk for miscarriage. A highly anticipated trial T4-LIFE that has not yet been published may further address the role of levothyroxine administration in women with TPOAb and recurrent miscarriage[29].

Sensitivity analyses based on timing of levothyroxine initiation demonstrated a reduction in preterm delivery when levothyroxine was initiated in the first trimester rather than preconception. However, this finding is likely attributed to the quality of the included studies since many of the trials that initiated levothyroxine in the first trimester were at high risk of bias and the funnel plot suggests publication bias. Furthermore, it appears biologically implausible that levothyroxine would be beneficial for thyroid autoimmune women when initiated during pregnancy but not when initiated preconception.

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A limitation of this study is that only one of the six included trials met all criteria for low risk of bias. Meta-analyses of live births and neonatal outcomes such as gestational age at delivery and birth weight were limited as only two trials reported these outcomes[15 16]. Furthermore, the sensitivity analysis is limited by the small number of studies included.

This study has many notable strengths. First, this study provides an updated summary of randomised controlled trials not included in previous systematic reviews and meta-analyses or available at the time of the American Thyroid Association 2017 pregnancy guideline recommendations development. The exclusion of observational studies minimizes bias and inclusion of only randomised controlled trials provides strong evidence to guide clinical decisions. Our systematic review and meta-analysis benefits from the use of rigorous standardized tools such as Cochrane Risk of Bias tool[14] a pre-specified analysis plan posted on Prospero, sensitivity analysis that address study quality and the assessment of publication bias.

## CONCLUSION

In summary, data from six randomised controlled trials fail to demonstrate evidence of benefit from levothyroxine treatment for pregnancy outcomes in thyroid autoimmune women with normal thyroid function and subclinical hypothyroidism. In the absence of overt thyroid dysfunction, current trial evidence does not support the levothyroxine treatment of thyroid autoimmunity preconception or during pregnancy. Given the lack of evidence of benefit of

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levothyroxine therapy, it is time to re-examine recommendations for screening TPOAb and treatment of women with thyroid autoimmunity preconception or in pregnancy.

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**Figure and Table Legend:**

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Flow Diagram

Figure 2. Risk of bias assessment and trial quality assessment.

Figure 3. Results of meta-analysis of effects of levothyroxine treatment on pregnancy and obstetrical outcomes. (A) pregnancy achieved, (B) miscarriage (C) preterm delivery (D) live birth

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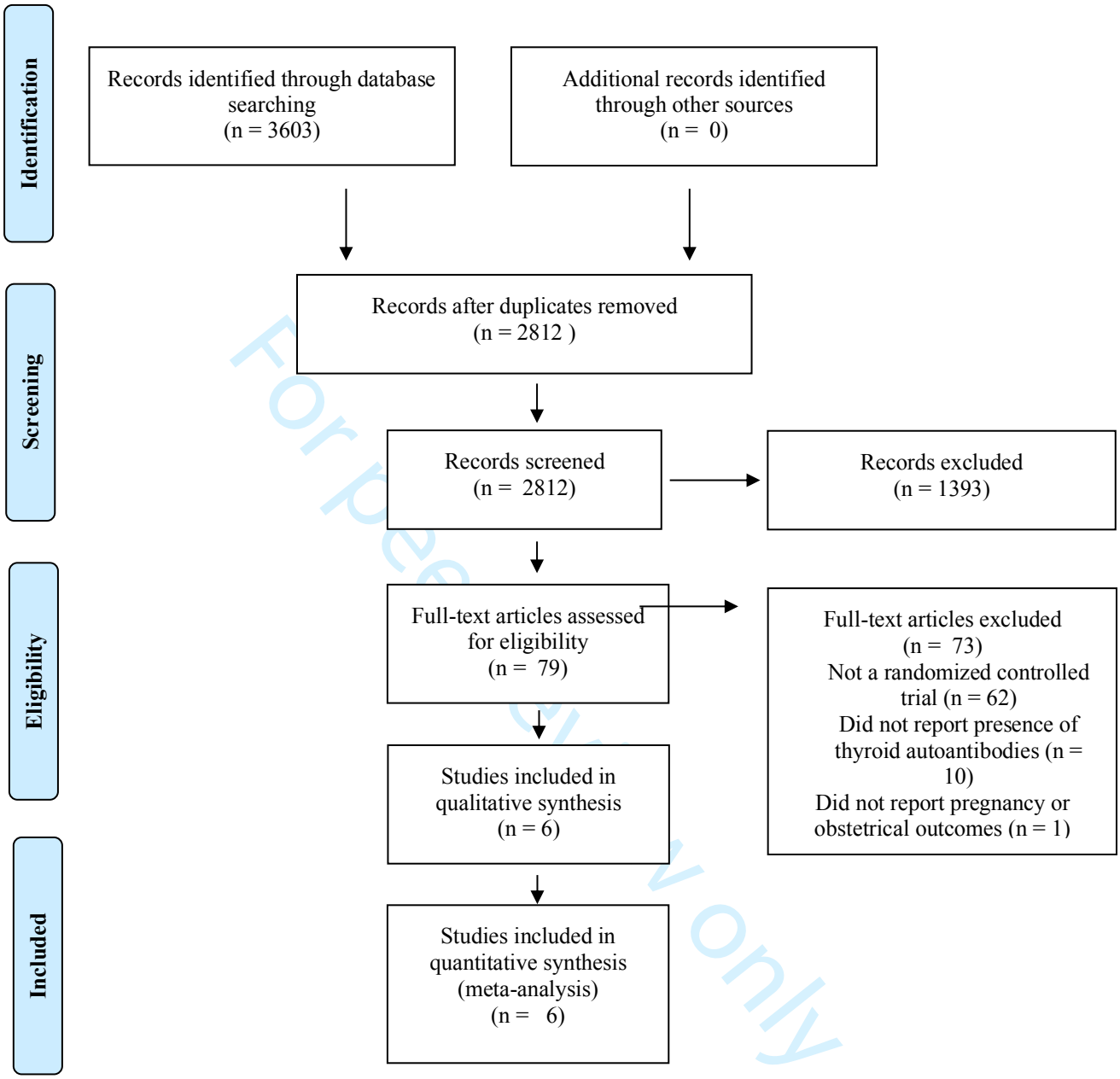



**Figure 1.** Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Flow Diagram

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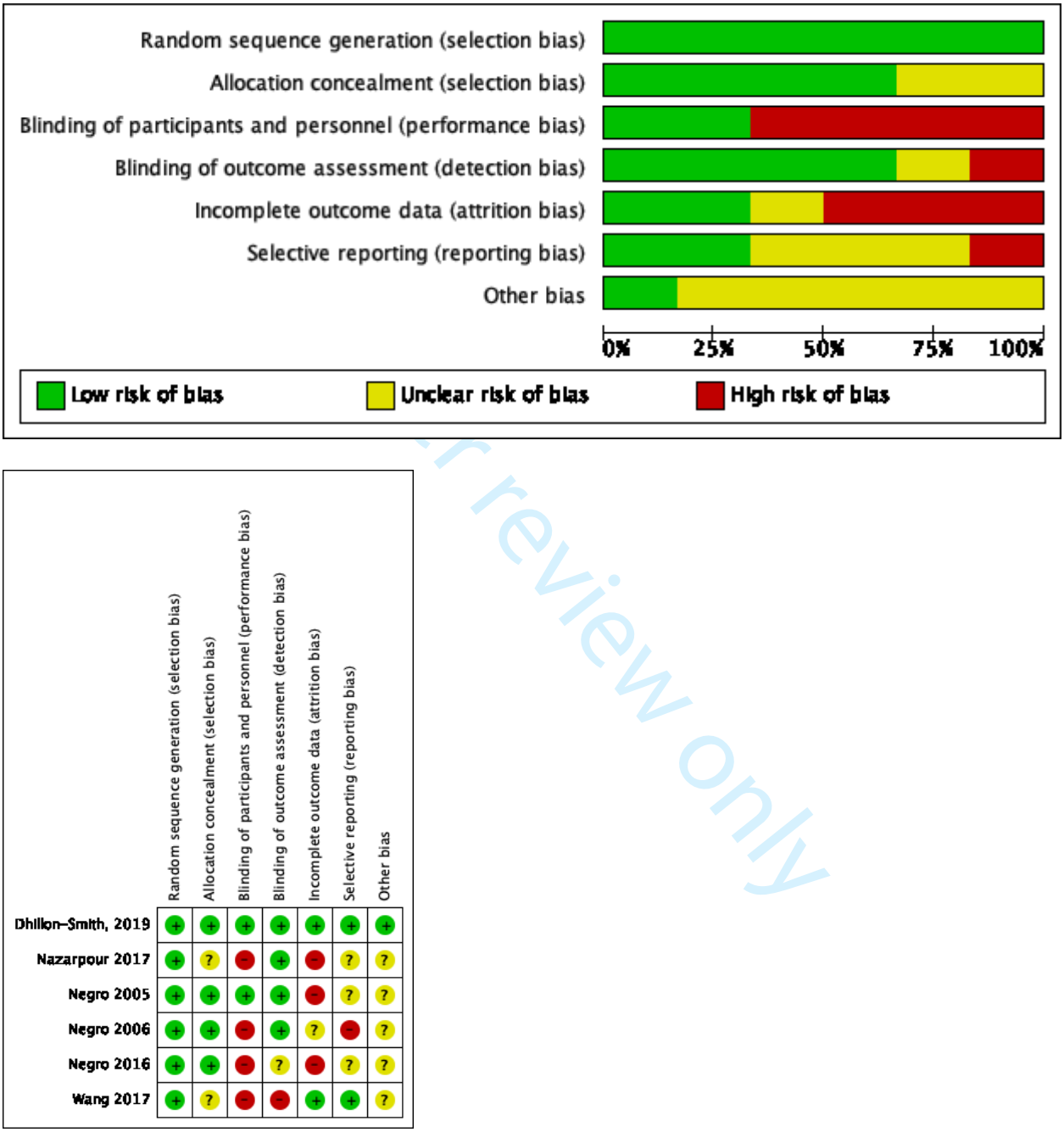
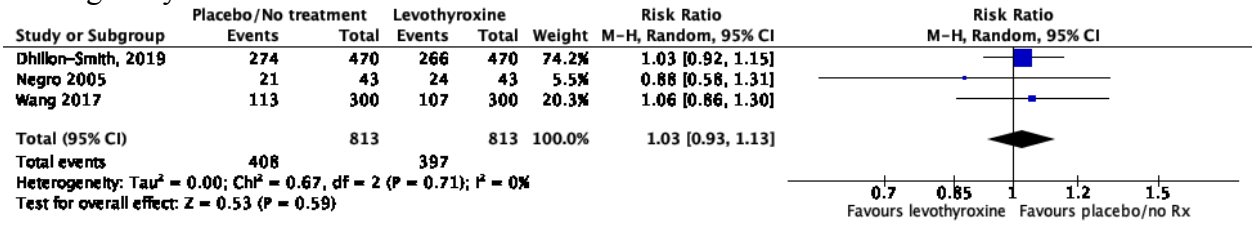


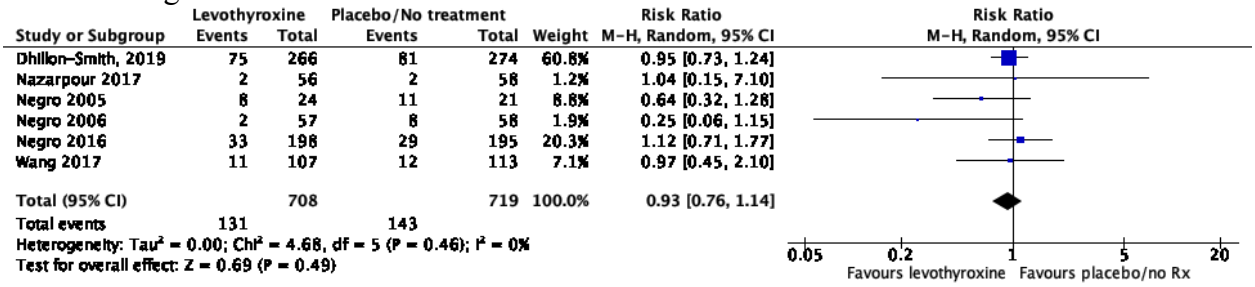

**Figure 2.** Risk of bias assessment and trial quality assessment.

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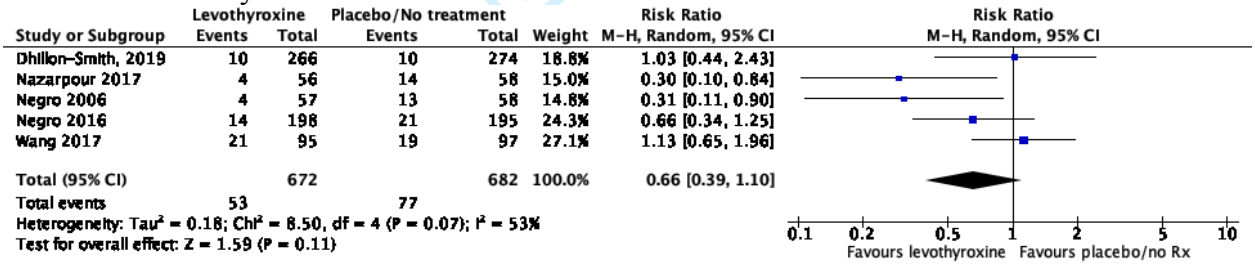
A. Pregnancy achieved




B. Miscarriage




C. Preterm delivery




D. Live births

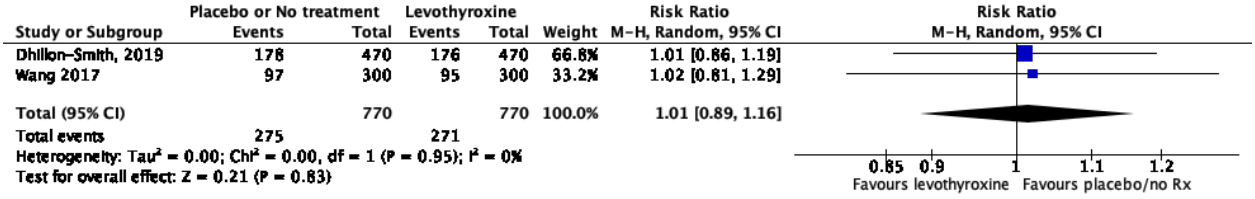



**Figure 3.** Meta-analysis of influence of levothyroxine on pregnancy outcomes. (A) pregnancy achieved, (B) miscarriage (C) preterm delivery (D) live birth

**Supplemental Information**

Supplemental Figure 1(S1): Full Search Strategy – MEDLINE (Ovid)

1. Hypothyroidism/
2. "hypothyroid*".ab,ti,kf.
3. "hypothyroxinemia".ab,ti,kf.
4. (("thyroid-stimulating hormone*" or "thyroid stimulating hormone*" or TSH*) adj2 deficien*).ab,ti,kf.
5. (thyroid* adj3 (autoimmun* or "auto-immun*" or antibod*)).ab,ti,kf.
6. Thyroxine/
7. "levothyro*".ab,ti,kf.
8. "thyrox*".ab,ti,kf.
9. synthroid.ab,ti,kf.
10. lt4.ab,ti,kf.
11. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10
12. exp Pregnancy/
13. "pregnan*".ab,ti,kf.
14. "gestat*".ab,ti,kf.
15. Pregnant Women/
16. "maternal*".ab,ti,kf.
17. "prenatal*".ab,ti,kf.
18. 12 or 13 or 14 or 15 or 16 or 17
19. 11 and 18
20. randomized controlled trial.pt.
21. controlled clinical trial.pt.
22. random*.ab.
23. placebo.ab.
24. drug therapy.fs.
25. trial.ab.
26. groups.ab.
27. 20 or 21 or 22 or 23 or 24 or 25 or 26
28. exp animals/ not humans.sh.
29. 27 not 28
30. 19 and 29
31. limit 30 to yr="1980 -Current"

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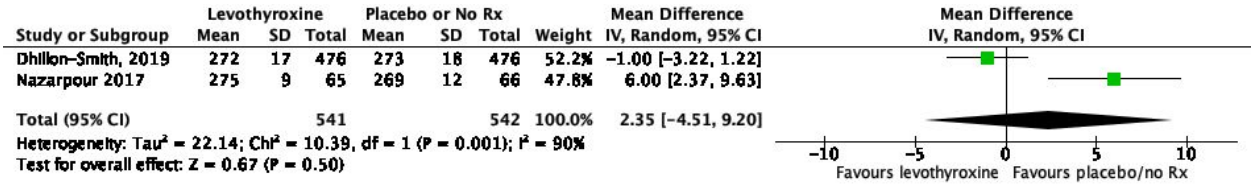
Supplemental Table (S2). Baseline patient characteristics

| | Maternal Age (years) | | Body Mass Index (kg/m$^2$) | | Baseline FT4 (pmol/L) | |
|---|---|---|---|---|---|---|
| **Study Author (year)** | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm |
| Dhillon-Smith *et al.* (2019) | 32.5 ± 4.9 | 32.7 ± 4.9 | 26.4± 5.6 | 26.5± 5.5 | 14.6±1.9 | 14.5±2 |
| Nazarpour *et al.* (2017) | 26.6 ± 5.8 | 27 ± 4.7 | 24.9± 5.1 | 24.6 ± 3.6 | | |
| Negro *et al.* (2005) | 29.2 ± 4 | 30.1 ± 5 | | | 11.2 ± 1.8 | 11.7 ± 0.7 |
| Negro *et al.* (2006) | 30 ± 5 | 30 ± 6 | | | 14.8 ± 4.2 | 14.6 ± 4.3 |
| Negro *et al.* (2016) | 28.9 ± 5.2 | 29.9 ± 5.1 | | | | |
| Wang *et al.* (2017) | 31.3 ± 3.9 | 31.7 ± 3.8 | 22.7 ± 3.3 | 22.8 ± 3.2 | | |

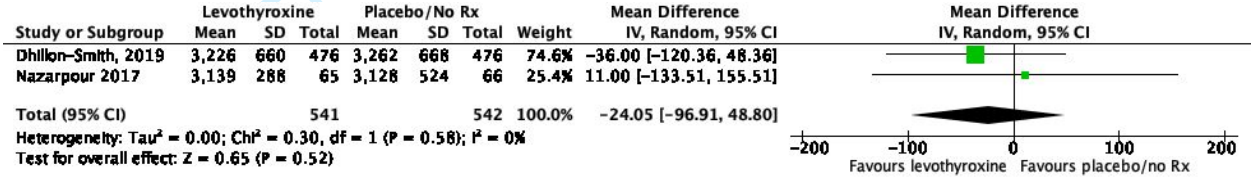
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Supplemental Figure (S3):

A. Weighted mean difference for gestational age at time of delivery (days)



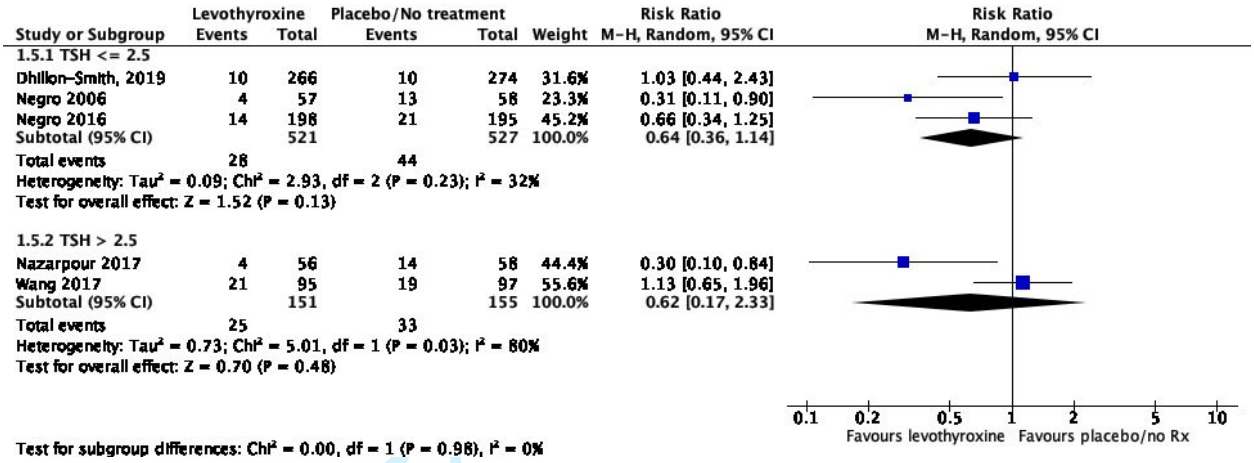

B. Weighted mean difference for birth weight (g)



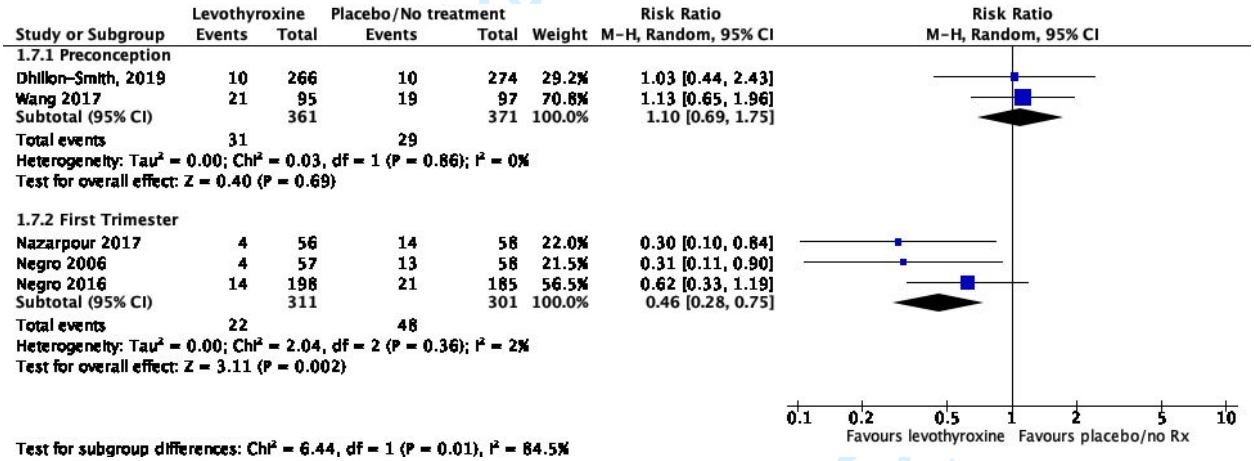

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Supplemental Figure (S4): Stratified meta-analysis for preterm delivery subgroups.

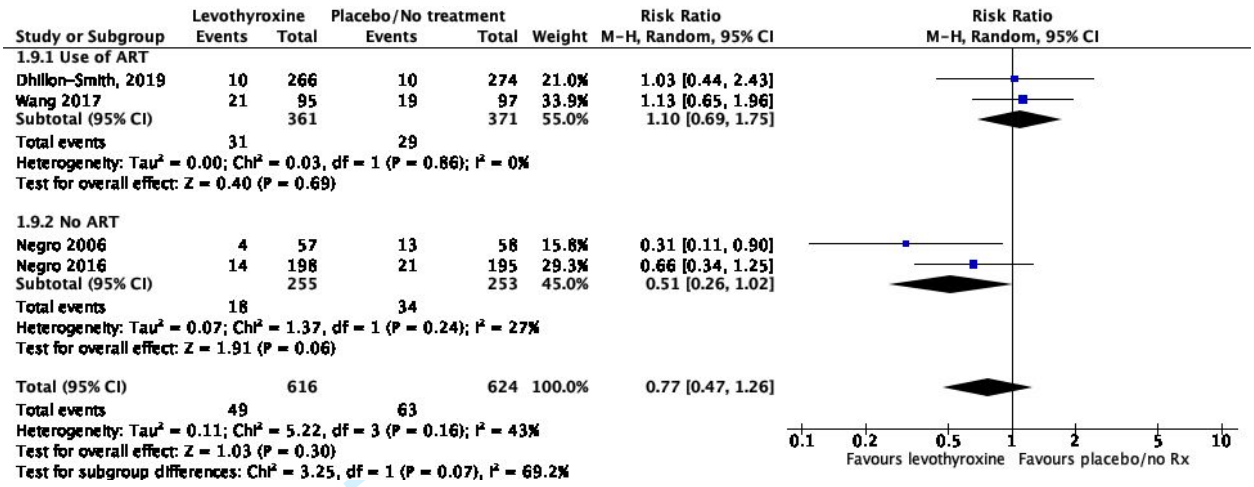
A. Stratified by TSH levels at baseline



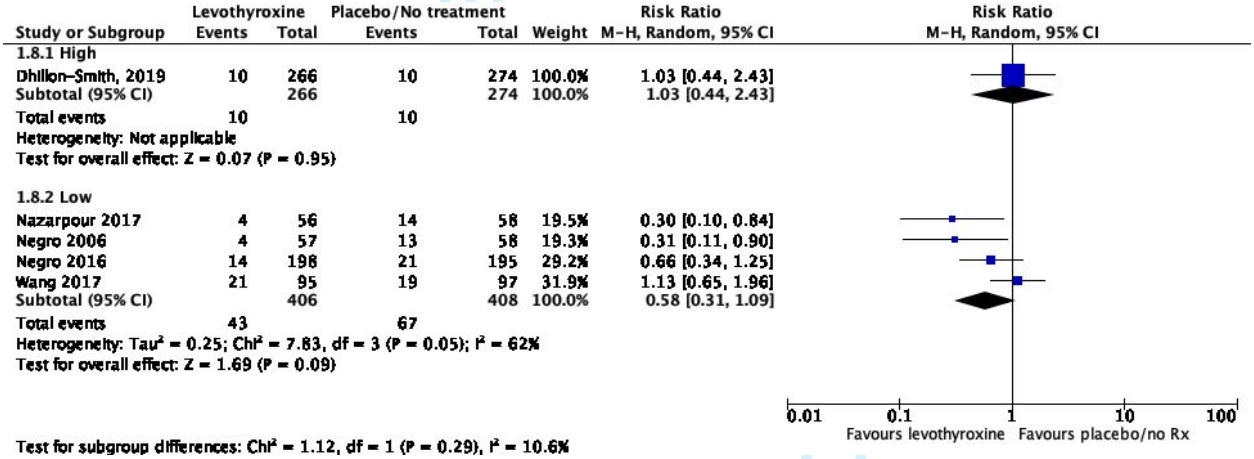

B. Stratified by timing of levothyroxine initiation



C. Stratified by use of assisted reproductive technologies

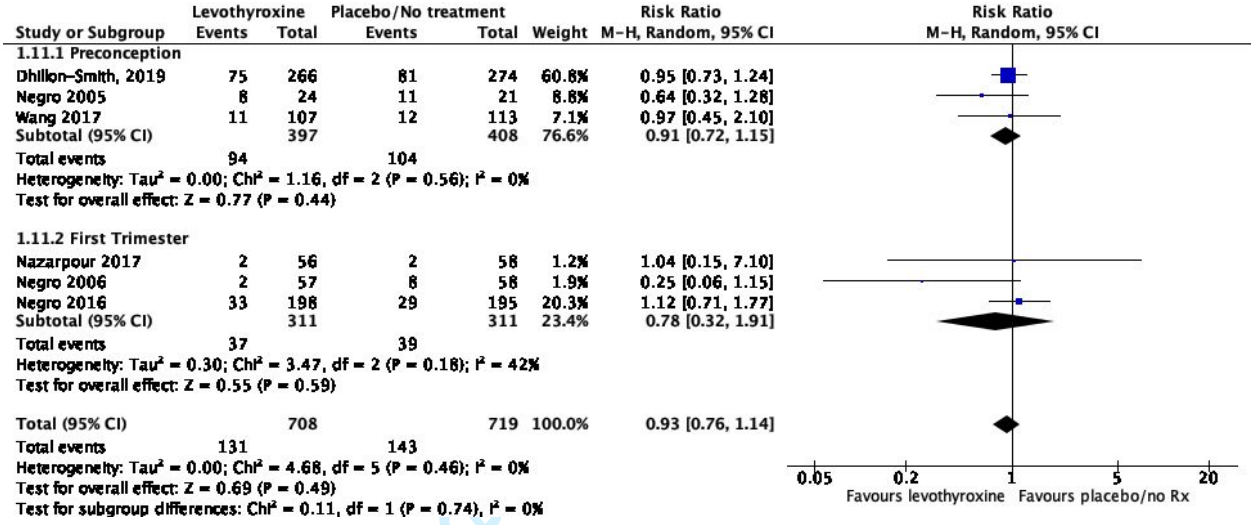



D. Stratified by study quality

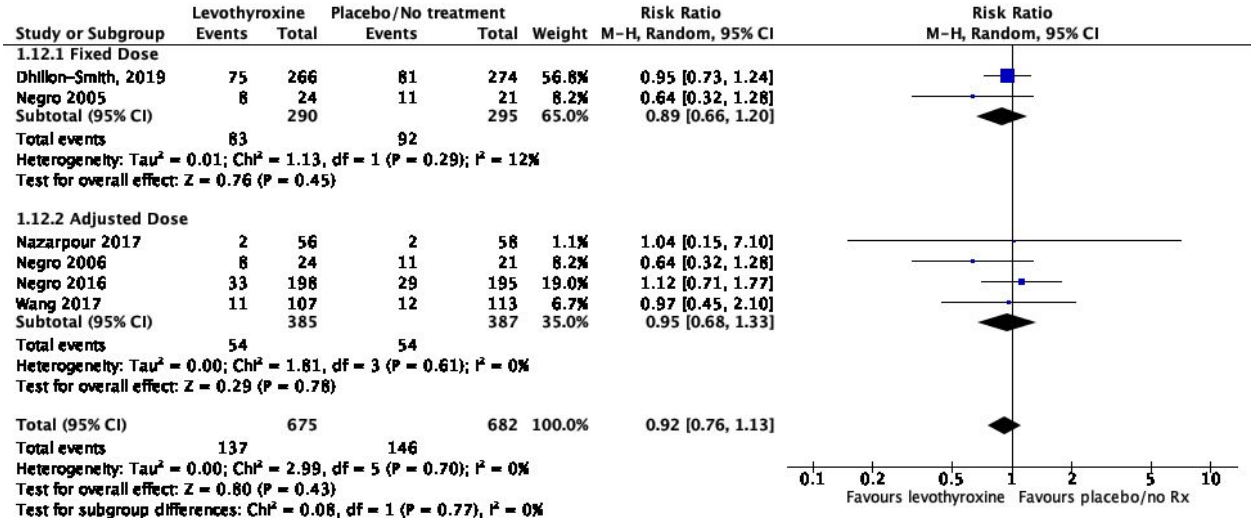


Supplemental Figure (S5): Stratified meta-analysis for miscarriage subgroups

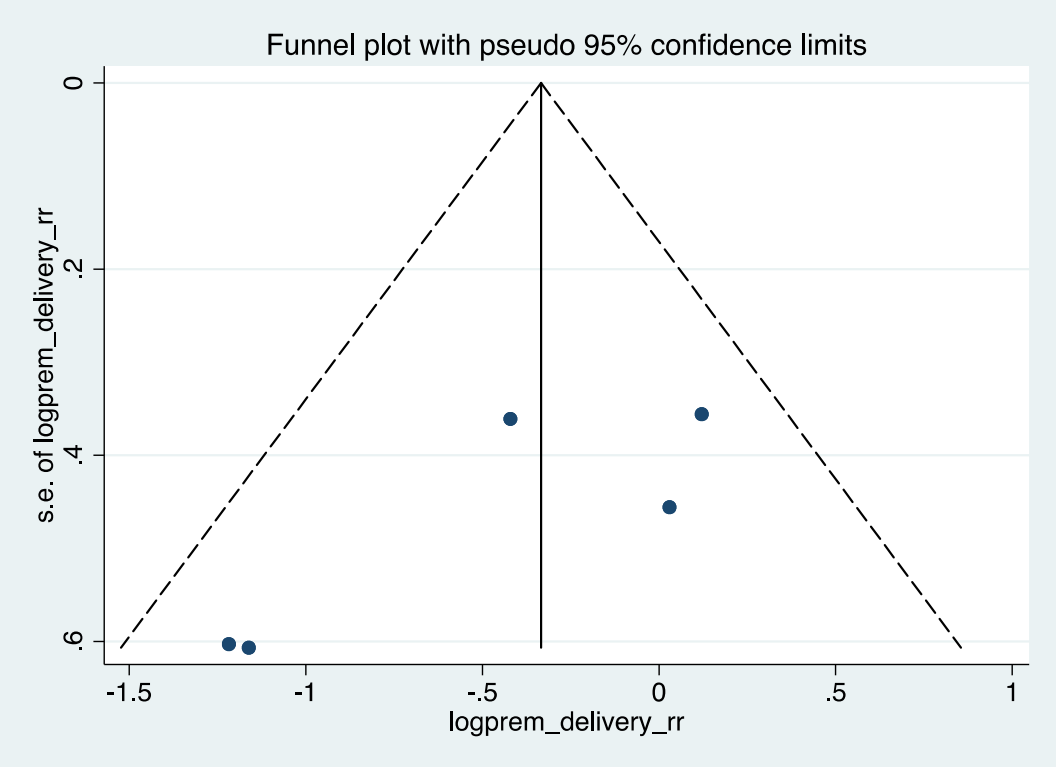
A. Stratified by timing of levothyroxine initiation




B. Stratified by fixed levothyroxine versus adjusted levothyroxine dose



Supplemental Figure (S6): Funnel plot for preterm delivery




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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 4 | Describe the rationale for the review in the context of what is already known. | 3-4 |
| Objectives | 4-5 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 5 |
| Eligibility criteria | 5 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 5 |
| Information sources | 5 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 5 |
| Search | 5 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 5-6 |
| Study selection | 6 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 6 |
| Data collection process | 6-7 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 6 |
| Data items | 7 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 6 |
| Risk of bias in individual studies | 7 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 7 |
| Summary measures | 7-8 | State the principal summary measures (e.g., risk ratio, difference in means). | 7-8 |
| Synthesis of results | 8 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 8 |

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# PRISMA 2009 Checklist



| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 7 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 7 |
| Additional analyses | 8 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 8 |
| **RESULTS** | | | |
| Study selection | Fig 1 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | Figure 1 |
| Study characteristics | Table 1 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | Table 1 |
| Risk of bias within studies | Fig 2 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | Figure 2 |
| Results of individual studies | Fig 3 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | Figure 3 |
| Synthesis of results | Fig 3 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | Figure 3 |
| Risk of bias across studies | Fig 2 | Present results of any assessment of risk of bias across studies (see Item 15). | Figure 2 |
| Additional analysis | Suppl | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | Appendix |
| **DISCUSSION** | | | |
| Summary of evidence | 13 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 13 |
| Limitations | 15-16 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 15 |
| Conclusions | 16 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 16 |
| **FUNDING** | | | |
| Funding | 3 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 1 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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# The Impact of Levothyroxine in Women with Positive Thyroid Antibodies on Pregnancy Outcomes: A Systematic Review and Meta-Analysis of Randomised Controlled Trials



| | |
|---|---|
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| Manuscript ID | bmjopen-2020-043751.R1 |
| Article Type: | Original research |
| Date Submitted by the Author: | 30-Nov-2020 |
| Complete List of Authors: | Lau, Lorraine; University of Calgary Cumming School of Medicine, Medicine<br>Benham, Jamie; University of Calgary Cumming School of Medicine, Medicine<br>Lemieux, Patricia; University of Calgary Cumming School of Medicine, Medicine<br>Yamamoto, Jennifer; University of Calgary Faculty of Medicine, Medicine<br>Donovan, Lois; University of Calgary Faculty of Medicine, Medicine |
| Primary Subject Heading: | Diabetes and endocrinology |
| Secondary Subject Heading: | Obstetrics and gynaecology |
| Keywords: | Thyroid disease < DIABETES & ENDOCRINOLOGY, Maternal medicine < OBSTETRICS, DIABETES & ENDOCRINOLOGY, OBSTETRICS |
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# The Impact of Levothyroxine in Women with Positive Thyroid Antibodies on Pregnancy Outcomes: A Systematic Review and Meta-Analysis of Randomised Controlled Trials

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## ABSTRACT

**Objective:** To evaluate the effect of levothyroxine therapy on pregnancy outcomes compared with placebo or no treatment in women without overt hypothyroidism who are TPOAb and/or TgAb-positive.

**Design:** Systematic review and meta-analysis of randomised controlled trials

**Study Eligibility Criteria**: Pre-specified criteria for inclusion were: randomised trials of levothyroxine versus control (placebo or no treatment) among women with positive TPOAb or TgAb who were pregnant or considering conception.

**Data Sources**: Ovid MEDLINE, EMBASE, CINAHL, Cochrane Database of Systematic Reviews, and Cochrane Central Register of Controlled Trials were searched from 1980 to November 5, 2020.

**Outcome Measures**: Pre-specified data elements were extracted and where appropriate, meta-analyses were conducted. Main outcomes include pregnancy achieved, miscarriage, preterm delivery and live birth.

**Risk of bias assessment:** Cochrane Risk of Bias Tool (Modified) for Quality Assessment of Randomised Controlled Trials.

**Results:** From 3023 citations, 79 citations were identified for full text review. Of these, six trials (total of 2,263 women) were included for qualitative and quantitative analyses. Risk of bias was deemed low for only one trial. There was no significant difference in the relative risk (RR) of pregnancy achieved (RR 1.03; 95% CI 0.93, 1.13), miscarriage (RR 0.93; 95% CI 0.76, 1.14), preterm delivery (RR 0.66; 95% CI 0.39, 1.10), or live births (RR 1.01; 95% CI 0.89, 1.16) in thyroid autoimmune women treated with levothyroxine compared to controls. Sensitivity analyses of preterm birth identified study quality and timing of levothyroxine initiation as sources of heterogeneity.

**Conclusions:** Among pregnant women or women planning conception, with thyroid autoimmunity, there is a lack of evidence of benefit for levothyroxine use (moderate to high GRADE). Recommendations to use levothyroxine in this setting need to be reconsidered.


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**Strengths and limitations of this study**

- This systematic review summarises the most current published randomised controlled trials not included in other systematic reviews and meta-analysis.
- This systematic review and meta-analysis is rigorous in design, methodology and analyses with a prepublished protocol.
- Of the six trials identified for inclusion, only one was deemed low risk of bias.
- There was evidence of publication bias.

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**Contribution to authorship:** PL and LED conceived the idea for this systematic review and meta-analysis, and conducted the literature search along with the initial title/abstract review. JMY contributed her expertise in systematic review methodology to the design and conduct of this systematic review. LWL and JLB conducted the full text review, study selection, data extraction, and statistical analyses. LWL wrote the first draft of the manuscript. All authors contributed to critical review and approval of the final manuscript.

**Ethics Statement.** The authors adhered to the Preferred Reporting Items for Systemic Reviews and Meta-Analyses (PRISMA) guidelines. Ethical committee approval was not required for this study.

**PROSPERO registration number** CRD42019130459

**Data Availability Statement:** All data relevant to the study are included in the article or uploaded as supplementary information.

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## INTRODUCTION:

As thyroid function is frequently screened during the early gestational period, identification of thyroid immunity, defined by the presence of thyroid peroxidase antibodies (TPOAb) and/or thyroglobulin antibodies (TgAb), is relatively common in pregnant women. The prevalence of TPOAb positivity in the first trimester ranges from 5 to 15% in unselected pregnant women[1 2]. The prevalence of thyroid autoimmunity is even higher in women with a history of recurrent pregnancy loss, reported as 23 to 31%[3 4].

The association between thyroid autoimmunity and poor pregnancy outcomes, specifically miscarriage and preterm birth, has been well described[5 6]. However, the mechanism linking thyroid autoimmunity with increased risk of miscarriage and preterm birth in women without overt hypothyroidism is unclear. Two competing mechanistic theories propose that 1) thyroid autoantibodies are a marker of a hostile immune environment, or 2) thyroid autoantibodies act as a surrogate biomarker of impaired thyroid hormone reserves, despite normal thyroid function tests[7]. If the latter theory is true, one could postulate that supplementation with levothyroxine may improve thyroid hormone levels and prevent negative pregnancy outcomes.

Whether or not levothyroxine therapy, either prior to or during pregnancy, improves pregnancy outcomes in women with thyroid autoimmunity is controversial[8-14]. The 2017 American Thyroid Association (ATA) pregnancy guidelines strongly recommends levothyroxine for women with thyroid autoimmunity and a thyroid stimulated hormone (TSH) above the pregnancy-specific

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reference range. These guidelines recommend a consideration of levothyroxine in thyroid autoimmune women with TSH above 2.5 mIU/L and among euthyroid women with positive TPOAb and a history of pregnancy loss[15]. The evidence evaluating levothyroxine treatment in pregnant women with thyroid autoantibodies is conflicting as evidenced by three recent systematic reviews and meta-analyses[13 16 17]. The discrepant results may be attributed to the inclusion of small observational studies and the lack of inclusion of a recently published large multi-center trial[8]. Additionally, the limitations of observational studies must be considered. Observational studies identify associations, but do not provide the highest level of evidence for determining causation or therapeutic efficacy. Randomised controlled trials are considered the gold standard to evaluate therapeutic efficacy[18]. Therefore, we performed a systematic review and meta-analysis to evaluate all randomised controlled trials of levothyroxine therapy in women with thyroid autoimmunity who were pregnant or planning pregnancy.

## METHODS

### Protocol and Registration

A systematic review and meta-analysis was performed as outlined in the registered protocol (PROSPERO CRD42019130459), and is reported in accordance with the Preferred Reporting Items for Reviews and Meta-Analyses (PRISMA).

### Data Sources and Search Strategy

The following databases were searched in duplicate on April 17, 2019 (and updated on November 5, 2020) from 1980: Ovid MEDLINE, EMBASE, CINAHL, Cochrane Database of

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Systematic Reviews, and Cochrane Central Register of Controlled Trials. Clinicaltrials.gov was searched for active trials. The search strategy was developed in consultation with a medical librarian and used a combination of keywords related to hypothyroidism or thyroid autoimmunity and pregnancy (Supplemental S1). The search was conducted using an adapted version of the Cochrane search filter for randomised controlled trials where possible and was limited to human studies[19].

**Study Selection**

Studies that met all the following criteria were included in this review: (1) pregnant women or women contemplating pregnancy (including patients seeking assisted reproductive technologies), with positive TPOAb or TgAb, euthyroid or with subclinical hypothyroidism (by any prespecified study definition); (2) randomised to levothyroxine versus control (placebo or no treatment); (3) reported pregnancy outcomes; (4) reported in either English or French; (5) published from 1980 to April 17, 2019. The publication year 1980 was specifically chosen to coincide with the availability of more sensitive TSH assays[20 21].

Study selection was completed in two separate stages by two independent reviewers (PL and LED or LWL and JLB). First, duplicates were removed, and titles and abstracts were screened for eligibility (by PL and LED). Second, full citations deemed eligible at the title and abstract stage were retrieved and reviewed for eligibility in a full-text format (by LWL and JLB). Inclusion criteria was applied to select eligible articles and reasons for exclusion were recorded. Agreement was recorded at each stage and reported as a kappa statistic[22]. All disagreements between reviewers were solved by consensus, and, if needed, by discussion with a third

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independent reviewer (LED). Prior to each selection step, a pilot calibration was conducted by the respective reviewers to review inclusion criteria. The reference list of included articles was searched for the addition of further relevant studies.

**Data Collection Process**

Relevant study information was extracted by two independent reviewers (LWL and JLB) using standardized data extraction forms in Microsoft Excel (Version 16.16.18, Microsoft Corporation, Redmon, WA, USA). Extracted data elements included: study design, country, participant inclusion and exclusion criteria, TPOAb and TgAb results, definition of TPOAb and TgAb positivity, TPOAb and TgAb assays used, thyroid stimulating hormone (TSH) and thyroxine (T4) assays used and normal ranges, baseline serum concentrations of TSH and T4, iodine status of the country, levothyroxine dosing, achieved TSH and T4 after treatment, gestational timing of levothyroxine initiation, duration of treatment, details of the intervention, quality of the methods, inclusion of patients with subclinical hypothyroidism and definition of subclinical hypothyroidism, patient baseline characteristics, number of patients included in each group, pregnancy outcomes and their definitions.

**Quality Assessment**

Cochrane Risk of Bias Tool (Modified) for Quality Assessment of Randomised Controlled Trials[23] was used to assess study quality and evaluate the risk of bias by two independent reviewers (LL and JB). This tool assesses seven methodologic domains and these components enable classification of randomised trials as “Low Risk of Bias”, “High Risk of Bias” or “Unclear Risk of Bias”.

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**Study Outcomes**

The following pregnancy outcomes were prespecified and included: pregnancy by trial definition, miscarriage, preterm delivery, live-birth, birth weight, large for gestational age, small for gestational age, gestational age at delivery, multiple pregnancy, stillbirth, neonatal intensive care unit admission, preeclampsia, and gestational diabetes. GRADE (Grading of Recommendations, Assessment, Development and Evaluations) was utilized for grading of quality of evidence for the main outcomes[24].

**Data Synthesis and Analysis**

Where appropriate, meta-analyses were conducted to evaluate the relative risk (RR) and 95% confidence intervals (95% CI) of pre-specified pregnancy outcomes including pregnancy achieved, miscarriage, preterm delivery and live-birth in participants randomised to levothyroxine therapy compared with placebo or no treatment. Random-effects models were used. Statistical heterogeneity was quantified using $I^2$ statistic, where $I^2$ >50% represents moderate and $I^2$ >75% represents substantial heterogeneity across studies[25]. The following prespecified sensitivity analyses were conducted to explore heterogeneity: 1) TSH levels at baseline, 2) time of levothyroxine initiation (preconception or during pregnancy), 3) use and type of assisted reproduction technologies, 4) maternal characteristics, and 5) study quality. Publication bias was assessed using a funnel plot as the graphical aid. Statistical analysis was performed using STATA version 14 (StataCorp, College Station, TX, USA) and Review Manager 5.3 (Version 5.3.5, The Cochrane Collaboration, Copenhagen, Denmark).

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## Patient and Public Involvement

No patient involved.

## RESULTS

The search results are summarised in Figure 1. Among the 3,023 citations reviewed for title and abstract, 79 citations were identified for full text review. Six randomised controlled trials (total of 2,263 women) met all eligibility criteria and were included in the qualitative and quantitative analyses[8-12 14]. There was a substantial level of agreement for both the title and abstract review ($\kappa$=0.64) and for the full text review ($\kappa$=0.70).

### Trial Characteristics

Trial characteristics are summarised in Table 1. Three trials initiated levothyroxine therapy prior to conception[8 11 14], whereas three trials initiated therapy in the first trimester[9 10 12]. Two trials compared levothyroxine to placebo[8 10], and the other four trials compared with no treatment[9 11 12 14]. Reproductive technologies were used in three trials[8 11 14], with one trial not specifying details with this regard. All trials included women with positive TPOAb with only one trial including TPO or Tg antibodies[14]. The TSH normal reference ranges and definition of TPO positivity were prespecified in each trial and are summarised in Table 1.

### Patient Characteristics

Baseline participant characteristics are summarised in Table 1 and Supplemental S2. Mean maternal age ranged from 26 years[9] to 32 years[8]. Mean reported body mass index ranged from 22.7 kg/m$^2$ [14] to 26.5 kg/m$^2$ [8]. Baseline TSH values varied according to assay specific reference

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ranges. Baseline free T4 values were comparable in the three trials that reported this information[8 11 12] (Supplemental S2). There was insufficient information to assess the outcome of childhood neurodevelopment.

**Risk of Bias Assessment**

Risk of bias assessment is presented in Figure 2. Only one trial[8] was considered "low risk" of bias in all seven domains. Four trials[8 10-12] were assessed at "low risk" for selection bias. Two trials[9 14] were "unclear risk of bias" in allocation concealment. In four trials comparing levothyroxine to no treatment[9 10 12 14], the assessors were not blinded to patient randomization and thus these trials were considered to be "high risk" for performance bias. Published trial protocols were located for only two trials[8 14]. Thus, only these two trials were assessed as "low risk" for reporting bias.

**Pregnancy Outcomes**

There were sufficient data for meta-analyses of four pregnancy outcomes, as summarised in Figure 3. These outcomes include pregnancy achieved, miscarriage, preterm delivery and live births.

Pregnancy achieved was reported in three trials[8 11 14] (n=1,626) that initiated levothyroxine pre-conception. There was no significant difference in pregnancy achieved in thyroid autoimmune women treated levothyroxine compared with control (RR 1.03 [95% CI 0.93, 1.13]; $I^2$=0%). GRADE was rated as high quality for this outcome.

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| | | | | | | | | | Mean GA of LT4 initiation (weeks) | | Baseline TSH (mIU/L) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Study Author (year)/ Country** | **N** | **Intervention levothyroxine** | **Control** | **Use of assisted reproduction technologies (% of participants)** | **Inclusion of subclinical hypothyroidism** | **TSH (Normal Range; mIU/L)** | **Defintion of TPOAb positivity (IU/mL)** | | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm |
| Dhillon-Smith *et al.* (2019)/ United Kingdom | 952 | Initiated preconception | Placebo | Yes (45%) | No | 0.44-3.63 | > 99% concordance with UK NEQAS IIA | | Preconception | Preconception | 2.1 | 2 |
| Nazarpour *et al.* (2017)/ Iran | 131 | Initiated 4-8 days following first prenatal visit | No treatment | Unclear | Yes | 0.1-2.5 | >50 | | 11.4 ± 4.2 | 11.4 ± 4.2 | 3.7 | 3.2 |
| Negro *et al.* (2005)/ Italy | 72 | Initiated 1 month before assisted reproduction technologies | No treatment | Yes (100%) | Yes | 0.27-3.2 | >100 | | Preconception | Preconception | 1.9 | 1.7 |
| Negro *et al.* (2006)/ Italy | 115 | Initiated at first endocrinological visit (2-7d after first obstetrics visit) | Placebo | No | No | 0.27-4.2 | >100 | | 10.4 ± 3.1 | 10.3 ± 3.1 | 1.6 | 1.7 |
| Negro *et al.* (2016)/ Italy | 393 | Initiated in first trimester prior to 12 weeks | No treatment | No | No | 0.5-2.5 | >16 | | 7.1 ± 1.2 | 7.0 ± 1.2 | 1.4 | 1.4 |
| Wang *et al.* (2017)/ China | 600 | Initiated 2-4 weeks prior to controlled ovarian hyperstimulation | No treatment | Yes (100%) | No | 0.45-4.78 | 60 | | Preconception | Preconception | 2.9 | 2.1 |

**Table 1.** Characteristics of included randomised controlled trials. N, sample size; TSH, thyroid stimulating hormone; TPOAb, thyroid peroxidase antibody; LT4, levothyroxine; GA, gestational age; FT4, free thyroxine

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Miscarriage was reported in all six trials (n=1,427) and was defined as pregnancy loss at less than 20 weeks[9], less than 24 weeks[8], less than 28wks (early miscarriage defined as first 12 weeks, late miscarriage defined as between 13-28 weeks)[14], or not clearly defined[10-12]. There was no significant difference in the relative risk (RR) of miscarriage in TPOAb-positive pregnant women treated with levothyroxine compared with control (RR 0.93 [95% CI 0.76, 1.14]; $I^2$=0%). Inclusion of trials[8 11 14] that initiated levothyroxine only preconception revealed no significant difference compared with control (RR 0.91 [95% CI 0.72, 1.15]; $I^2$=0%). GRADE was rated as high quality for this outcome.

Preterm delivery was reported in five trials (n= 1,354)[8-10 12 14]. Preterm delivery was defined as live birth prior to 34 weeks[8] or 37 weeks[9 10 12 14]. There was no significant difference in the relative risk of preterm delivery in thyroid autoimmune pregnant women treated with levothyroxine compared with control (RR 0.66 [95% CI 0.39, 1.10]; $I^2$=53%). GRADE was rated at moderate quality for this outcome. Downgrade considerations include considerable inconsistency, wide confidence intervals contributing to imprecision and evidence of publication bias.

Live births were reported in two trials (n= 1,540)[8 14]. Live birth was defined as delivery of a living fetus beyond 27 weeks[14] or beyond 34 weeks[8]. There was no significant difference in live births in thyroid autoimmune women treated with levothyroxine compared with control (RR 1.01 [95% CI 0.89, 1.16]; $I^2$ = 0%). GRADE was rated at high quality for this outcome.

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Only one trial[10] reported on pre-eclampsia and found no difference in this outcome among thyroid autoimmune women treated with levothyroxine compared with control. None of the studies reported the outcome of gestational diabetes.

**Neonatal Outcomes:**

Neonatal outcomes were only reported in two trials (n = 1,083)[8 9]. Meta-analysis for gestational age at time of delivery (measured in days) and birth weight (measured in grams) are summarised in Supplemental S3. The weighted mean difference for gestational age at time of delivery in women treated with levothyroxine compared with placebo or no treatment was 2.35 days (95% CI -4.51, 9.20; $I^2 = 90\%$). Weighted mean difference for birth weight in women treated with levothyroxine compared with placebo or no treatment was -24.05g (95% CI -96.91, 48.80; $I^2 = 0\%$).

Only one trial[9] reported neonatal intensive care unit admission. They found fewer admissions among thyroid autoimmune women treated with levothyroxine (2 of 56 compared with control 12 of 58, $p < 0.05$). No studies reported large for gestational age, small for gestational age or multiple pregnancy.

**Sensitivity Analyses**

A sensitivity analysis was performed to explore the heterogeneity of our meta-analysis for preterm delivery. Results are summarised in Supplemental S4.

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TSH levels at baseline were subdivided into TSH ≤ 2.5 mIU/mL[8 10 12] and TSH > 2.5 mIU/mL[9 14]. The relative risk of preterm delivery in pregnant women with baseline TSH ≤ 2.5 mIU/mL was 0.64 (95% CI 0.36, 1.14; $I^2$=32%) compared with a relative risk of 0.62 (95% CI 0.17, 2.33; $I^2$=80%) in pregnant women with baseline TSH > 2.5 mIU/mL.

Timing of levothyroxine initiation was stratified into preconception[8 14] and in pregnancy[9 10 12]. The relative risk of preterm delivery in women initiated on levothyroxine preconception was 1.10 (95% CI 0.69, 1.75; $I^2$=0%) compared a relative risk of 0.46 (95% CI 0.28, 0.75; $I^2$=2%) in women initiated on levothyroxine in the first trimester.

The relative risk of preterm delivery in women using assisted reproductive technologies (ART)[8 14] was 1.10 (95% CI 0.69, 1.75, $I^2$ = 0%) compared with a relative risk of 0.51 (95% CI 0.26, 1.02; $I^2$ = 27%) in women that did not use ART[10 12].

Sensitivity analyses was also performed to explore the impact of fixed[8 11] and adjusted levothyroxine[9 10 12 14] doses on miscarriage rates (Supplemental S5). The relative risk of fixed levothyroxine dose was 0.89 (95% CI 0.66, 1.20; $I^2$=12%) compared with adjusted levothyroxine dose of 0.95 (95% CI 0.68, 1.33; $I^2$ = 0%).

Study quality was defined as high if there was low risk of bias across all seven domains of the Cochrane Collaboration's tool. We were unable to perform a sensitivity analysis exploring the impact of risk of bias on preterm delivery, as only one trial met the definition for high quality per the Cochrane Collaboration tool[8]. A funnel plot for preterm delivery is shown in Supplemental

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S6. On visual inspection, there was asymmetry indicating potential publication bias towards smaller studies reporting positive findings.

## DISCUSSION

This systematic review and meta-analysis of randomised controlled trials of levothyroxine therapy in women with thyroid autoimmunity without overt thyroid dysfunction demonstrates no evidence of benefit of levothyroxine therapy across prespecified pregnancy outcomes. Specifically, treatment with levothyroxine did not improve pregnancies achieved, miscarriage preterm delivery or live-birth across a total of 2,263 women, inclusive of those with a history of pregnancy loss and use of reproductive technologies. Furthermore, the availability of larger and better-quality trials since the publications of the American Thyroid Association 2017 pregnancy recommendations signals the need to update these guidelines as they pertain to the treatment of thyroid autoimmune women.

Small randomised clinical trials initially demonstrated that levothyroxine therapy reduced miscarriage and/or pre-term delivery in TPOAb-positive women compared with placebo or no treatment intervention[9-11]. However, the unclear definition of miscarriage and pre-term delivery among these early trials means that selective reporting of outcomes could have biased their reported findings. Subsequent large, multi-center randomised controlled trials with clear prespecified outcome definitions have shown no impact of levothyroxine therapy on maternal or fetal outcomes compared with no levothyroxine treatment in pregnant women with subclinical hypothyroidism, isolated hypothyroxinemia or positive TPOAb[8 14 26 27]. These intervention trials

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involved pregnant women or women seeking pregnancy, including individuals with a history of pregnancy loss.

A systematic review and meta-analysis of a large study population (n = 47,045) of individual participant and population data demonstrated increased risk of preterm birth in TPOAb-positive women[6]. In light of these findings, Cappola and Casey[28] highlighted the need to address whether levothyroxine treatment decreases the risk of preterm birth among women with TPOAb. Three other systematic reviews and meta-analyses examined the effect of levothyroxine supplementation in pregnant women with thyroid autoimmunity[17 29 30]. Unlike our review which only included randomised controlled trials, two of the other reviews[24,25] included observational studies. The findings of our meta-analysis were consistent with the review by Sun *et al*.[17], which showed no benefit of levothyroxine supplementation on their pregnancy outcomes. In contrast, Rao *et al.*[29 30] demonstrated a beneficial effect of levothyroxine treatment on reducing pregnancy loss and preterm birth among pregnant women with subclinical hypothyroidism and/or thyroid autoimmunity. Several notable distinctions likely account for these discrepant results. First, the Dhillon-Smith *et al.*[8] publication was not included in these reviews. Furthermore, both reviews by Rao *et al.*[29 30] included a trial that strongly influenced their results but did not report outcome data in sufficient detail to be included in our systematic review[31]. Our attempts to clarify the outcome data of this trial[31] with the authors were left unresponded.

The findings of our study are consistent with the results of a recent systematic review and meta-analysis of randomised controlled trials of women with positive TPO antibodies [32]. These authors also conclude that levothyroxine supplementation is not associated with improvement in

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pregnancy outcomes. This further emphasizes the need to reconsider the current 2017 ATA guidelines on use of levothyroxine in euthyroid women with thyroid autoimmunity.

Most miscarriages occur early in pregnancy, prior to the first obstetrics appointment, thus initiation of levothyroxine prior to conception best addresses its impact on miscarriage rates. The pathophysiology underlying the association between thyroid autoimmunity in euthyroid pregnant women and miscarriage is still not completely understood. One hypothesis postulates that thyroid autoimmunity reflects inadequate thyroid reserves to accommodate for increased physiologic demands of pregnancy[33]. It has been postulated that delayed initiation of levothyroxine in pregnancy may account for its failure to reduce miscarriage. Our meta-analysis found that initiation of levothyroxine preconception or in the first trimester did not impact miscarriage outcomes. This suggests that inadequate thyroid reserve in women with thyroid autoimmunity does not account for the increased risk for miscarriage. A highly anticipated trial T4-LIFE that has not yet been published may further address the role of levothyroxine administration in women with TPOAb and recurrent miscarriage[34].

Sensitivity analyses based on timing of levothyroxine initiation demonstrated a reduction in preterm delivery when levothyroxine was initiated in the first trimester rather than preconception. However, this finding is likely attributed to the quality of the included studies since many of the trials that initiated levothyroxine in the first trimester were at high risk of bias and the funnel plot suggests publication bias. Furthermore, it appears biologically implausible that levothyroxine would be beneficial for thyroid autoimmune women when initiated during pregnancy but not when initiated preconception.

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A limitation of this study is that only one of the six included trials met all criteria for low risk of bias. Meta-analyses of live births and neonatal outcomes such as gestational age at delivery and birth weight were limited as only two trials reported these outcomes[8 9]. Furthermore, the sensitivity analysis is limited by the small number of studies included. Lastly, the funnel plot suggests publication bias towards smaller studies reporting positive findings. Interpretation of the funnel plot is limited by the small number of studies included in this meta-analysis[35].

This study has many notable strengths. First, this study provides an updated summary of randomised controlled trials not included in previous systematic reviews and meta-analyses or available at the time of the American Thyroid Association 2017 pregnancy guideline recommendations development. The systematic review and meta-analysis provides moderate (GRADE) evidence for the preterm delivery and high (GRADE) evidence for the outcomes of pregnancy achieved, miscarriage and live birth to guide clinical decisions. Our systematic review and meta-analysis benefits from the use of rigorous standardized tools such as Cochrane Risk of Bias tool[23] a pre-specified analysis plan posted on Prospero, sensitivity analysis that address study quality and the assessment of publication bias.

## CONCLUSION

In summary, data from six randomised controlled trials fail to demonstrate evidence of benefit from levothyroxine treatment for pregnancy outcomes in thyroid autoimmune women with normal thyroid function and subclinical hypothyroidism. In the absence of overt thyroid

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dysfunction, current trial evidence does not support the levothyroxine treatment of thyroid autoimmunity preconception or during pregnancy. Given the lack of evidence of benefit of levothyroxine therapy, it is time to re-examine recommendations for screening TPOAb and treatment of women with thyroid autoimmunity preconception or in pregnancy.

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**Figure and Table Legend:**

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Flow Diagram

Figure 2. Risk of bias assessment and trial quality assessment.

Figure 3. Results of meta-analysis of effects of levothyroxine treatment on pregnancy and obstetrical outcomes. (A) pregnancy achieved, (B) miscarriage (C) preterm delivery (D) live birth

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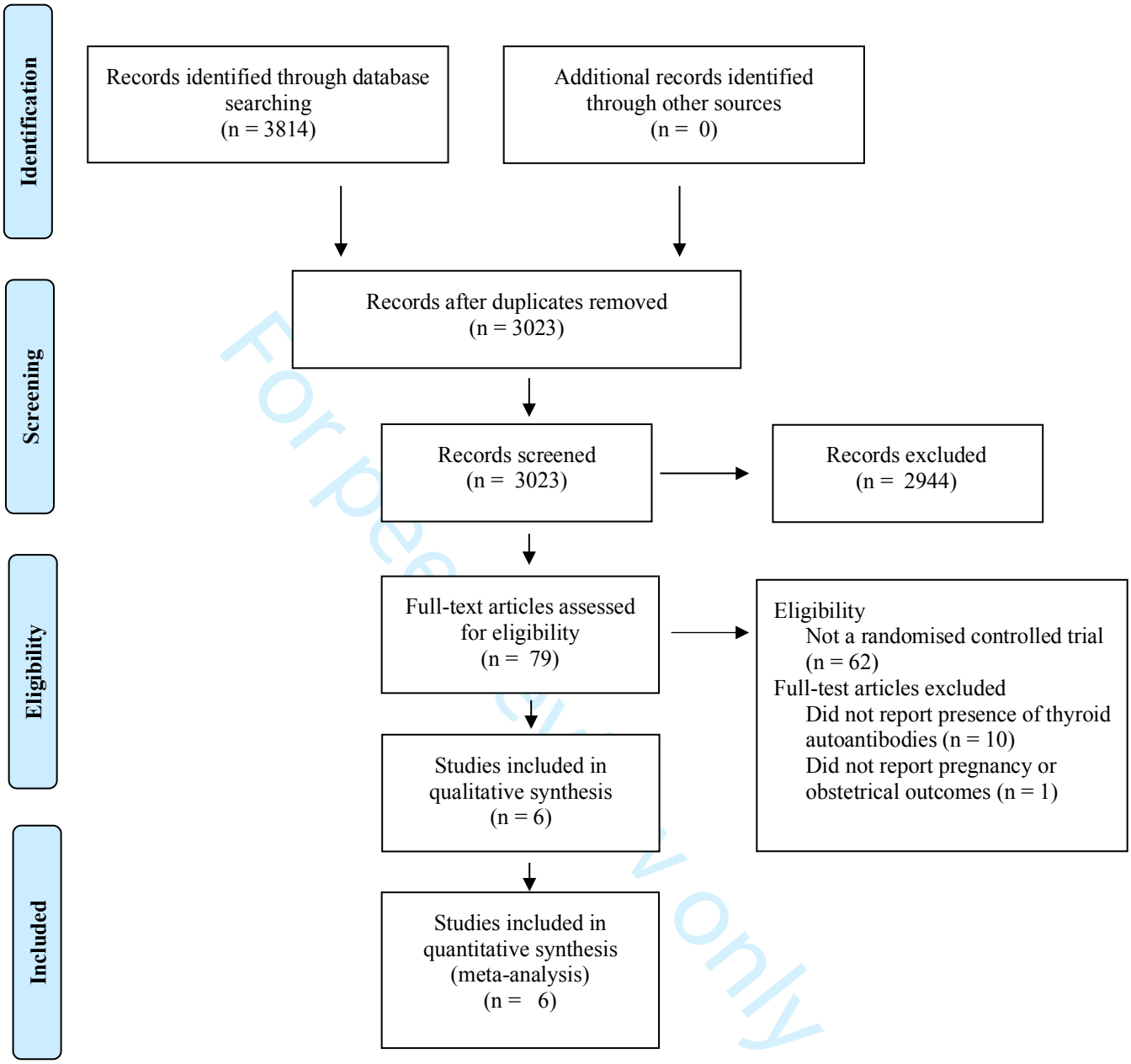



**Figure 1.** Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Flow Diagram

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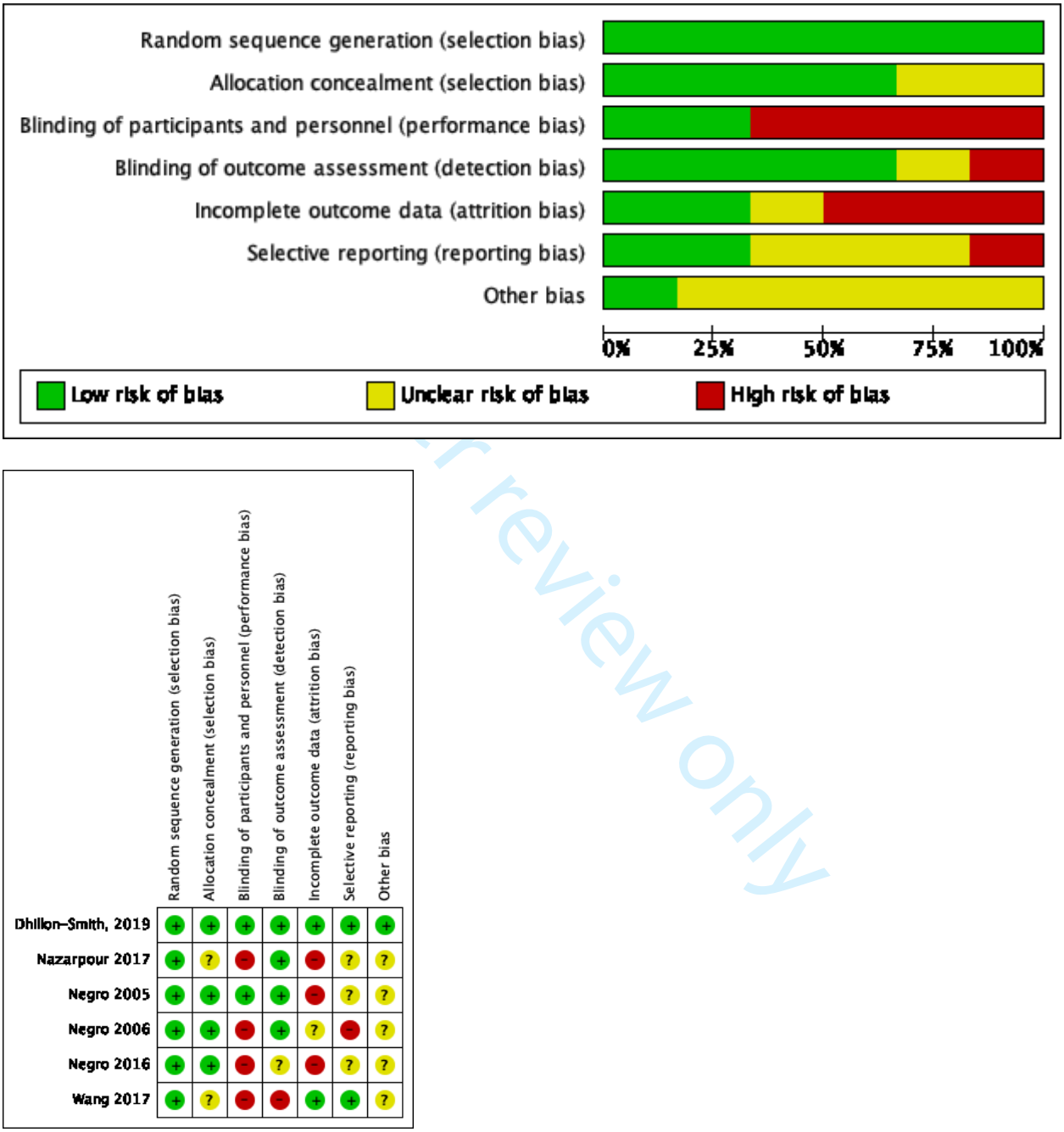
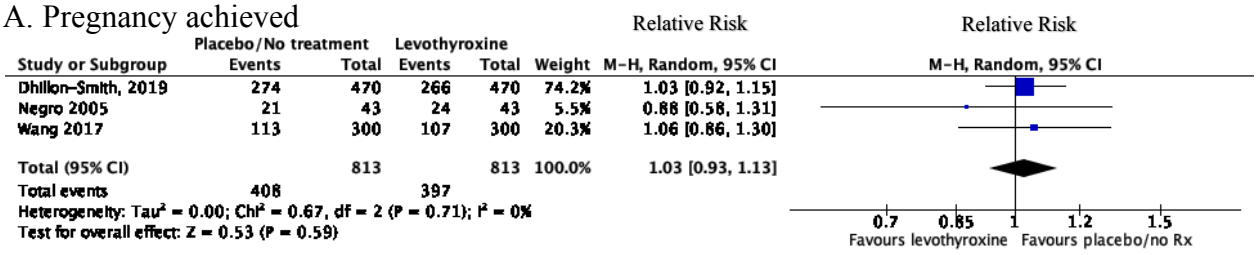


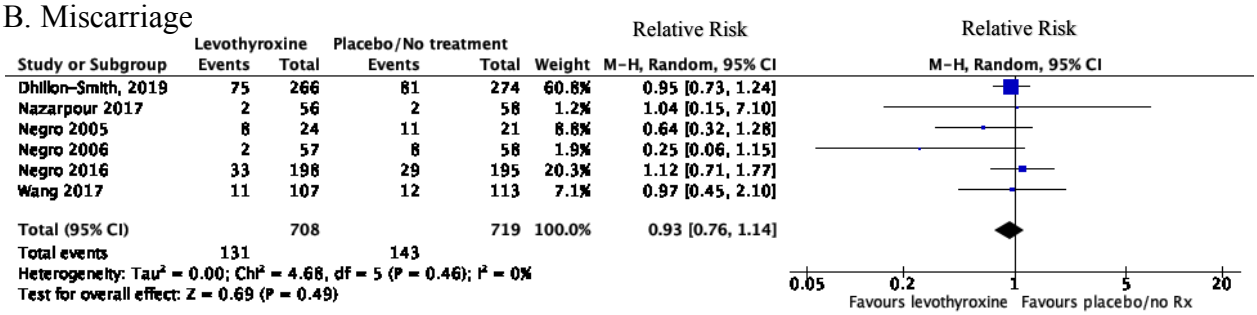

**Figure 2.** Risk of bias assessment and trial quality assessment.

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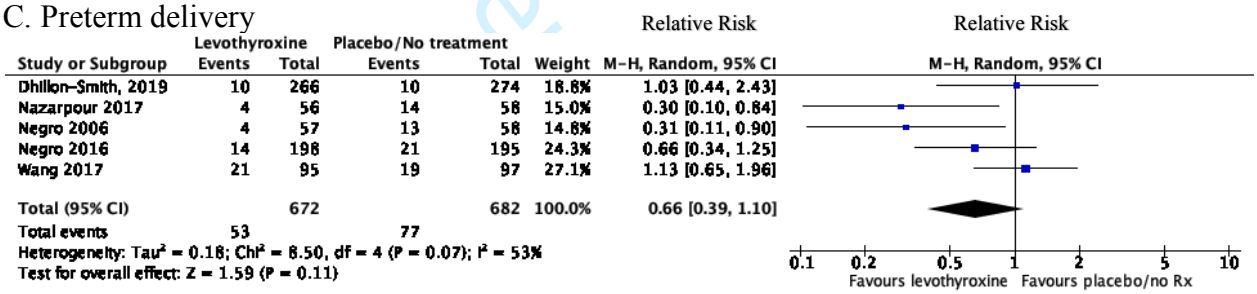
A. Pregnancy achieved




B. Miscarriage




C. Preterm delivery




D. Live births

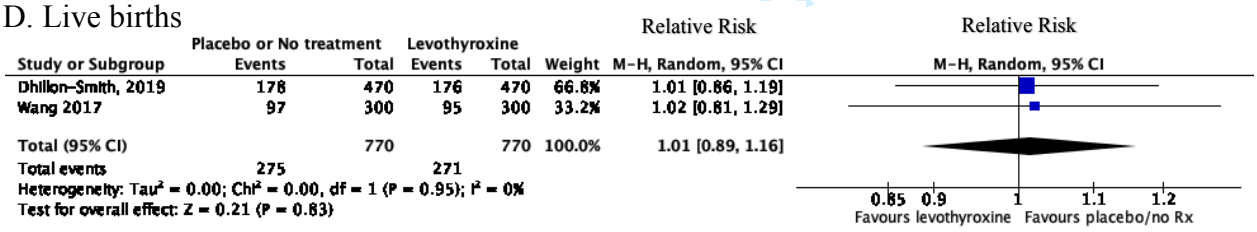



**Figure 3.** Meta-analysis of influence of levothyroxine on pregnancy outcomes. (A) pregnancy achieved, (B) miscarriage (C) preterm delivery (D) live birth

**Supplemental Information**

Supplemental Figure 1(S1): Full Search Strategy – MEDLINE (Ovid)

1. Hypothyroidism/
2. "hypothyroid*".ab,ti,kf.
3. "hypothyroxinemia".ab,ti,kf.
4. (("thyroid-stimulating hormone*" or "thyroid stimulating hormone*" or TSH*) adj2 deficien*).ab,ti,kf.
5. (thyroid* adj3 (autoimmun* or "auto-immun*" or antibod*)).ab,ti,kf.
6. Thyroxine/
7. "levothyro*".ab,ti,kf.
8. "thyrox*".ab,ti,kf.
9. synthroid.ab,ti,kf.
10. lt4.ab,ti,kf.
11. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10
12. exp Pregnancy/
13. "pregnan*".ab,ti,kf.
14. "gestat*".ab,ti,kf.
15. Pregnant Women/
16. "maternal*".ab,ti,kf.
17. "prenatal*".ab,ti,kf.
18. 12 or 13 or 14 or 15 or 16 or 17
19. 11 and 18
20. randomized controlled trial.pt.
21. controlled clinical trial.pt.
22. random*.ab.
23. placebo.ab.
24. drug therapy.fs.
25. trial.ab.
26. groups.ab.
27. 20 or 21 or 22 or 23 or 24 or 25 or 26
28. exp animals/ not humans.sh.
29. 27 not 28
30. 19 and 29
31. limit 30 to yr="1980 -Current"

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Supplemental Table (S2). Baseline patient characteristics

| | Maternal Age (years) | | Body Mass Index (kg/m$^2$) | | Baseline FT4 (pmol/L) | | Duration of Treatment | LT4 dosing and titration | Funding source |
|---|---|---|---|---|---|---|---|---|---|
| **Study Author (year)** | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm | | | |
| Dhillon-Smith *et al.* (2019) | 32.5 ± 4.9 | 32.7 ± 4.9 | 26.4± 5.6 | 26.5± 5.5 | 14.6±1.9 | 14.5±2 | Until the end of pregnancy | LT4 was d/c if TFT were outside assay specific limits at 6-8 wks, 1-18wks and 28 wks | UK national institute for health research |
| Nazarpour *et al.* (2017) | 26.6 ± 5.8 | 27 ± 4.7 | 24.9± 5.1 | 24.6 ± 3.6 | N/A | N/A | Throughout pregnancy | If TSH < 1.0 mcIU/mL: 0.5mcg/kg/d; If TSH 1.0-2.0, 0.75mcg/kg/d; If TSH > 2.0 or TPOAb > 1500IU/mL then 1mcg/kg/day | Did not receive any specific grant from any funding agency |
| Negro *et al.* (2005) | 29.2 ± 4 | 30.1 ± 5 | N/A | N/A | 11.2 ± 1.8 | 11.7 ± 0.7 | Throughout pregnancy | 1mg/kg/day | Unclear |
| Negro *et al.* (2006) | 30 ± 5 | 30 ± 6 | N/A | N/A | 14.8 ± 4.2 | 14.6 ± 4.3 | Throughout pregnancy | If TSH < 1.0 mcIU/mL: 0.5mcg/kg/d; If TSH 1.0-2.0, 0.75mcg/kg/d; If TSH > 2.0 or TPOAb > 1500IU/mL then 1mcg/kg/day | Unclear |
| Negro *et al.* (2016) | 28.9 ± 5.2 | 29.9 ± 5.1 | N/A | N/A | N/A | N/A | Throughout pregnancy | If TSH 0.5-1.5, LT4 0.5mcg/kg/d; TSH 1.5-2.5, LT4 1mcg/kg/d. Titration: If TSH > 3.0 in 2nd or 3rd trimester, LT4 dose | Unclear |

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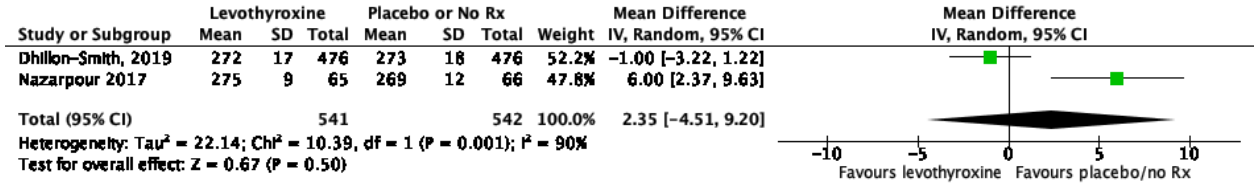
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | increased by 12.5mcg/d; If TSH <0.5 in 2nd or 3rd trimester, then decrease by 12.5mcg/d | |
| Wang *et al.* (2017) | 31.3 ± 3.9 | 31.7 ± 3.8 | 22.7 ± 3.3 | 22.8 ± 3.2 | N/A | N/A | Throughout pregnancy | STARTING Dose: TSH >/= 2.5mIU/L: starting dose is 50mcg/d; TSH <2.5mIU/L, starting dose is 25mcg/d. If body weight <50kg, then starting dosing decreased by 50%<br><br>Titration: TSH in first trimester 0.1-2.5mIU/L, 0.2-3.0mIU/L in second trimester, 0.3-3.0mIU/L in third trimester | National Key Technology R and D program and Chinese National 973 program (Both form the Ministry of Science and Technology of China) |

Table S2: N/A specifies that information was not available. FT4 denotes free thyroxine. LT4 denotes levothyroxine.

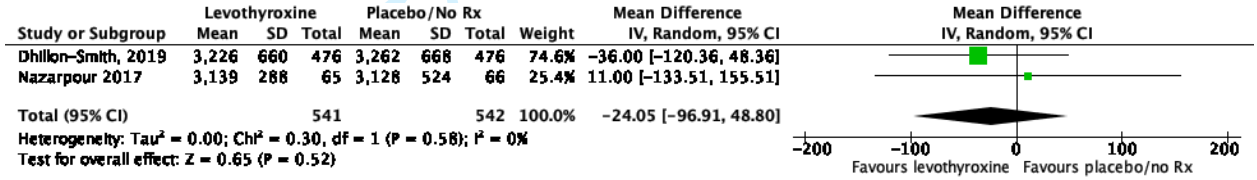
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Supplemental Figure (S3):

A. Weighted mean difference for gestational age at time of delivery (days)



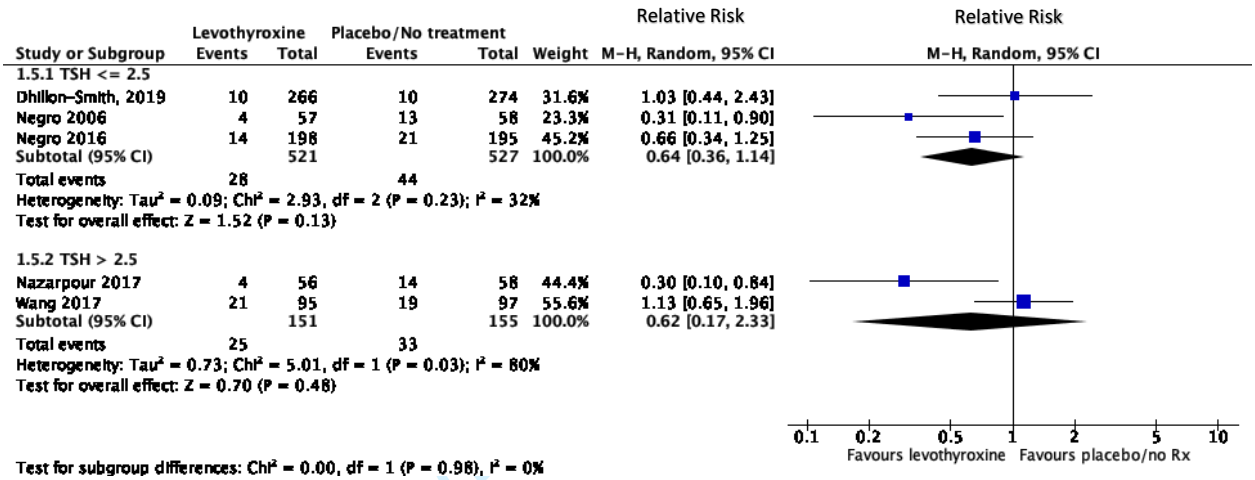

B. Weighted mean difference for birth weight (g)



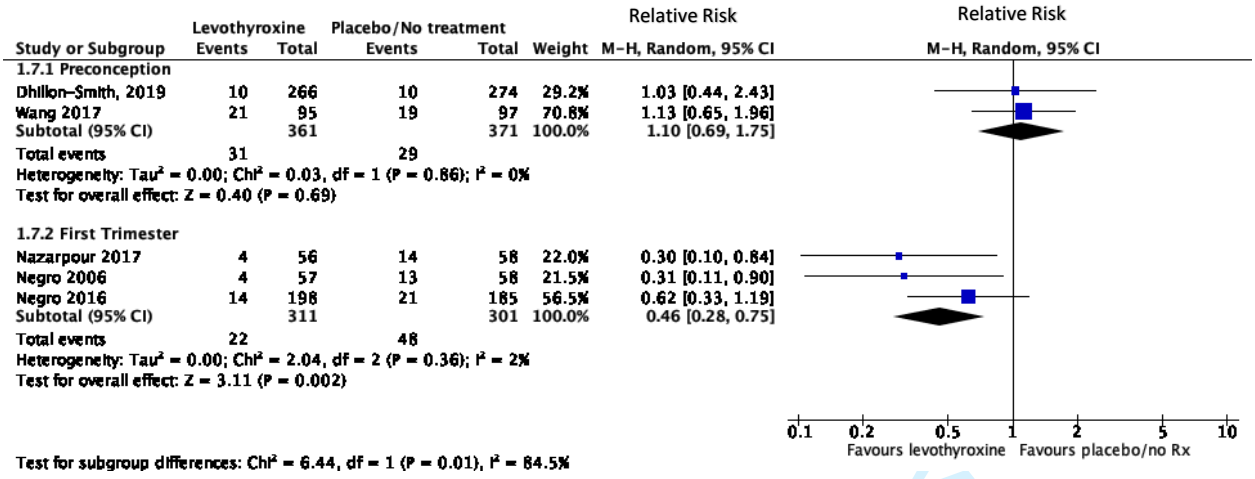

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Supplemental Figure (S4): Stratified meta-analysis for preterm delivery subgroups.

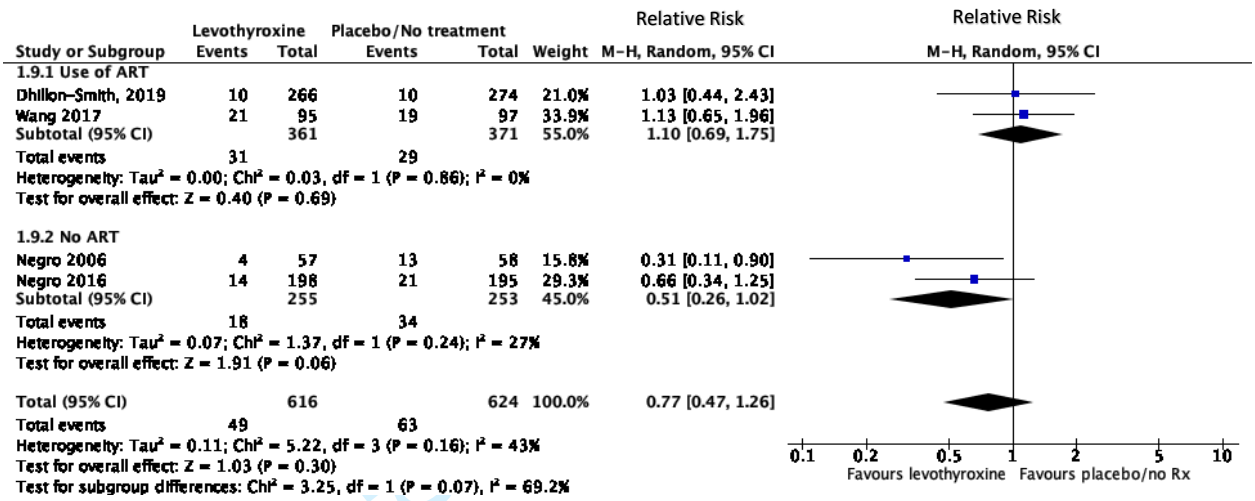
A. Stratified by TSH levels at baseline



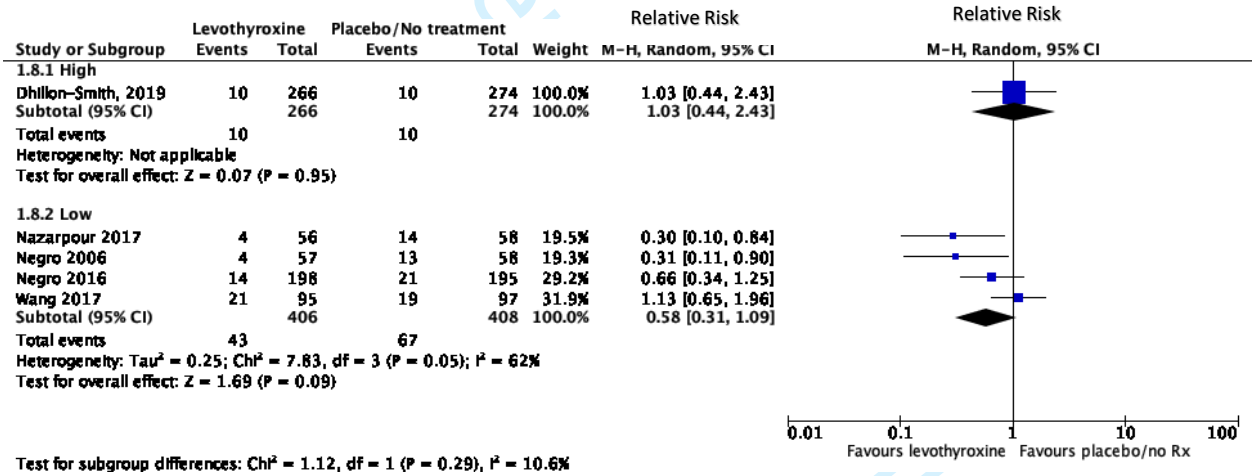

B. Stratified by timing of levothyroxine initiation



C. Stratified by use of assisted reproductive technologies

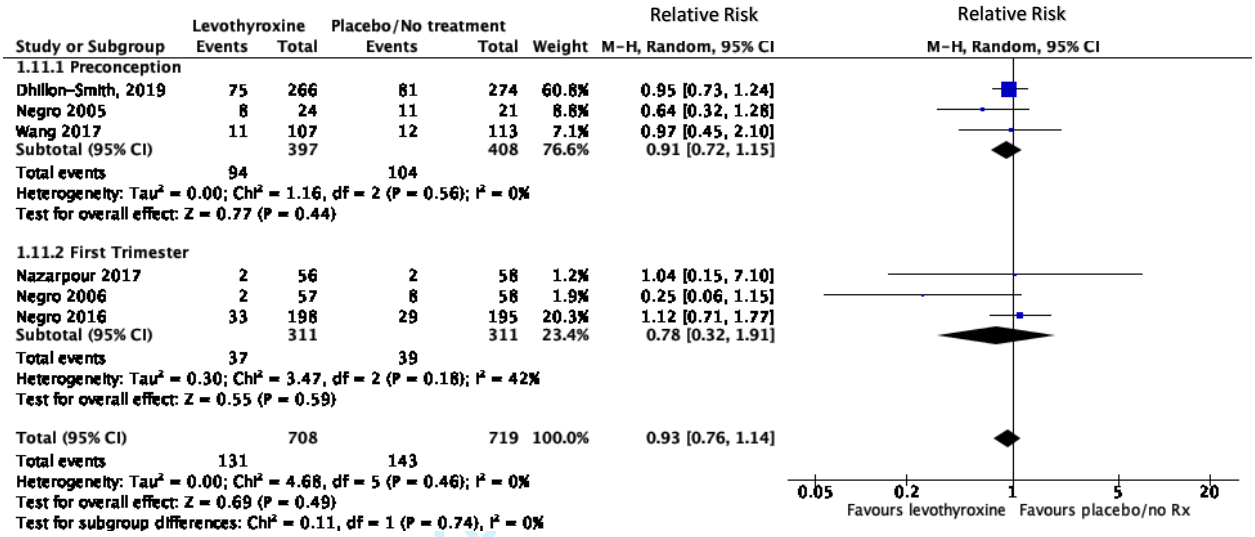



D. Stratified by study quality

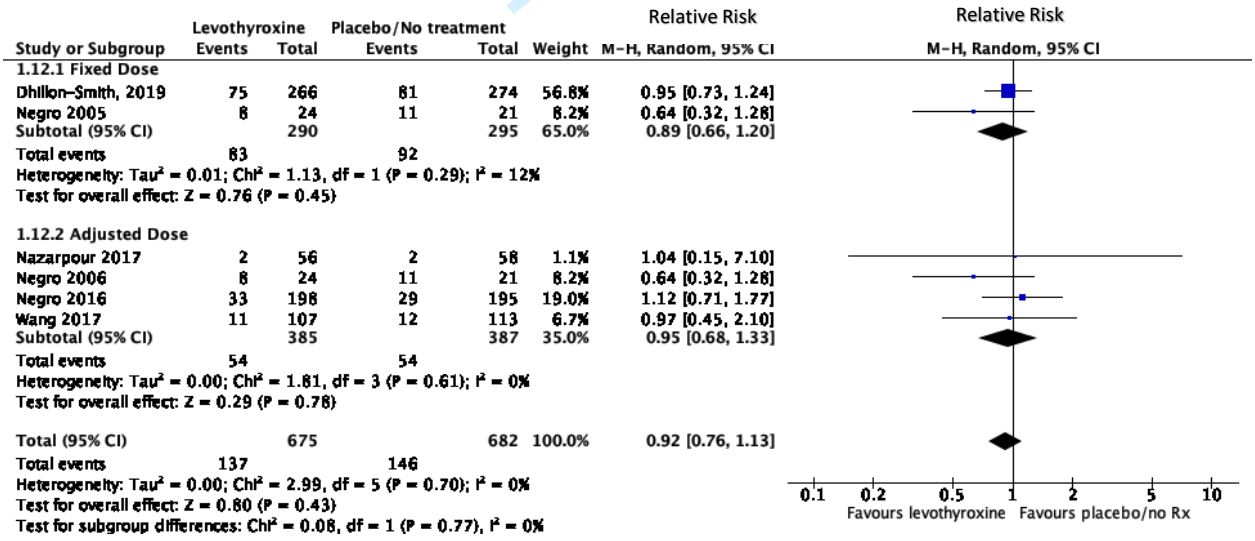


Supplemental Figure (S5): Stratified meta-analysis for miscarriage subgroups

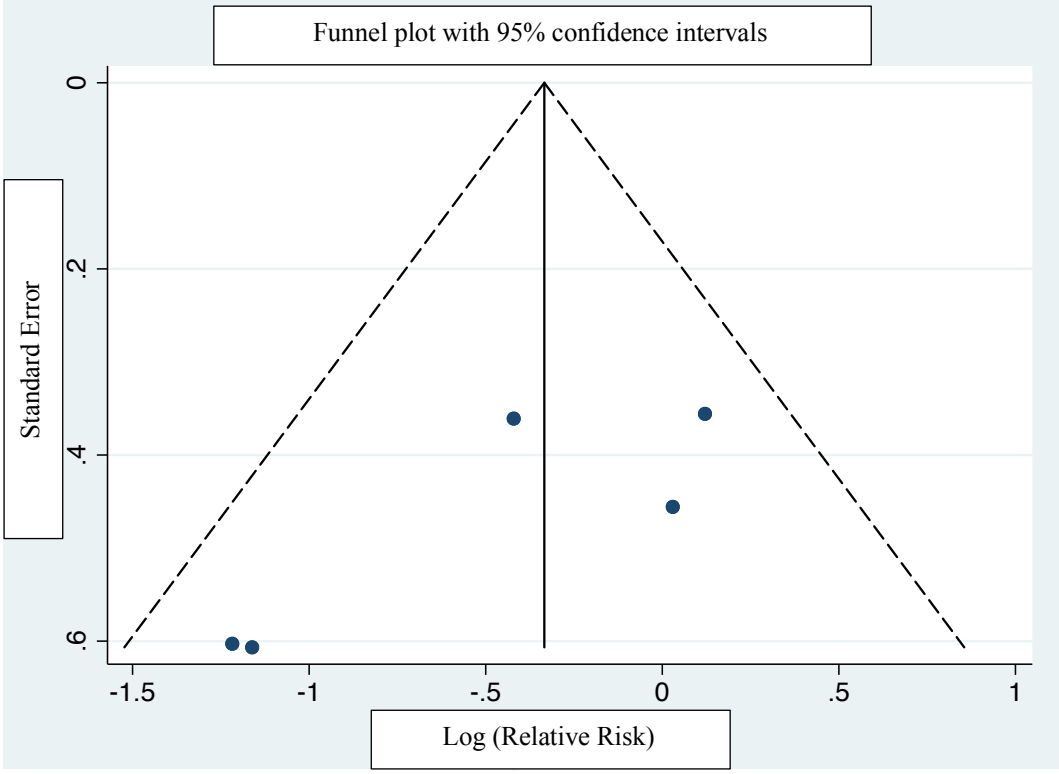
A. Stratified by timing of levothyroxine initiation




B. Stratified by fixed levothyroxine versus adjusted levothyroxine dose




Supplemental Figure (S6): Funnel plot for preterm delivery


Funnel plot with 95% confidence intervals
Standard Error
0
.2
.4
.6
-1.5
-1
-.5
0
.5
1
Log (Relative Risk)

review only

# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 4 | Describe the rationale for the review in the context of what is already known. | 3-4 |
| Objectives | 4-5 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 5 |
| Eligibility criteria | 5 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 5 |
| Information sources | 5 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 5 |
| Search | 5 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 5-6 |
| Study selection | 6 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 6 |
| Data collection process | 6-7 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 6 |
| Data items | 7 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 6 |
| Risk of bias in individual studies | 7 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 7 |
| Summary measures | 7-8 | State the principal summary measures (e.g., risk ratio, difference in means). | 7-8 |
| Synthesis of results | 8 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 8 |

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# PRISMA 2009 Checklist



| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 7 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 7 |
| Additional analyses | 8 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 8 |
| **RESULTS** | | | |
| Study selection | Fig 1 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | Figure 1 |
| Study characteristics | Table 1 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | Table 1 |
| Risk of bias within studies | Fig 2 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | Figure 2 |
| Results of individual studies | Fig 3 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | Figure 3 |
| Synthesis of results | Fig 3 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | Figure 3 |
| Risk of bias across studies | Fig 2 | Present results of any assessment of risk of bias across studies (see Item 15). | Figure 2 |
| Additional analysis | Suppl | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | Appendix |
| **DISCUSSION** | | | |
| Summary of evidence | 13 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 13 |
| Limitations | 15-16 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 15 |
| Conclusions | 16 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 16 |
| **FUNDING** | | | |
| Funding | 3 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 1 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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# The Impact of Levothyroxine in Women with Positive Thyroid Antibodies on Pregnancy Outcomes: A Systematic Review and Meta-Analysis of Randomised Controlled Trials



| | |
|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2020-043751.R2 |
| Article Type: | Original research |
| Date Submitted by the Author: | 19-Dec-2020 |
| Complete List of Authors: | Lau, Lorraine; University of Calgary Cumming School of Medicine, Medicine<br>Benham, Jamie; University of Calgary Cumming School of Medicine, Medicine<br>Lemieux, Patricia; University of Calgary Cumming School of Medicine, Medicine<br>Yamamoto, Jennifer; University of Calgary Faculty of Medicine, Medicine<br>Donovan, Lois; University of Calgary Faculty of Medicine, Medicine |
| Primary Subject Heading: | Diabetes and endocrinology |
| Secondary Subject Heading: | Obstetrics and gynaecology |
| Keywords: | Thyroid disease < DIABETES & ENDOCRINOLOGY, Maternal medicine < OBSTETRICS, DIABETES & ENDOCRINOLOGY, OBSTETRICS |
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# The Impact of Levothyroxine in Women with Positive Thyroid Antibodies on Pregnancy Outcomes: A Systematic Review and Meta-Analysis of Randomised Controlled Trials

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## ABSTRACT

**Objective:** To evaluate the effect of levothyroxine therapy on pregnancy outcomes compared with placebo or no treatment in women without overt hypothyroidism with presence of thyroid peroxidase antibodies (TPOAb) and/or thyroglobulin antibodies (TgAb).

**Design:** Systematic review and meta-analysis of randomised controlled trials

**Study Eligibility Criteria**: Pre-specified criteria for inclusion were: randomised trials of levothyroxine versus control (placebo or no treatment) among women with positive TPOAb or TgAb who were pregnant or considering conception.

**Data Sources**: Ovid MEDLINE, EMBASE, CINAHL, Cochrane Database of Systematic Reviews, and Cochrane Central Register of Controlled Trials were searched from 1980 to November 5, 2020.

**Outcome Measures**: Pre-specified data elements were extracted and where appropriate, meta-analyses were conducted. Main outcomes include pregnancy achieved, miscarriage, preterm delivery and live birth.

**Risk of bias assessment:** Cochrane Risk of Bias Tool for Quality Assessment of Randomised Controlled Trials.

**Results:** From 3023 citations, 79 citations were identified for full text review. Of these, six trials (total of 2,263 women) were included for qualitative and quantitative analyses. Risk of bias was deemed low for only one trial. There was no significant difference in the relative risk (RR) of pregnancy achieved (RR 1.03; 95% CI 0.93, 1.13), miscarriage (RR 0.93; 95% CI 0.76, 1.14), preterm delivery (RR 0.66; 95% CI 0.39, 1.10), or live births (RR 1.01; 95% CI 0.89, 1.16) in thyroid autoimmune women treated with levothyroxine compared to controls. Sensitivity analyses of preterm birth identified study quality and timing of levothyroxine initiation as sources of heterogeneity.

**Conclusions:** Among pregnant women or women planning conception, with thyroid autoimmunity, there is a lack of evidence of benefit for levothyroxine use (moderate to high GRADE). Recommendations to use levothyroxine in this setting need to be reconsidered.


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**Strengths and limitations of this study**

- This systematic review summarises the most current published randomised controlled trials not included in other systematic reviews and meta-analysis.
- This systematic review and meta-analysis is rigorous in design, methodology and analyses with a prepublished protocol.
- Of the six trials identified for inclusion, only one was deemed low risk of bias.
- There was evidence of publication bias.

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**Competing interests statement:** The authors have no competing interests to declare.

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**Contribution to authorship:** PL and LED conceived the idea for this systematic review and meta-analysis, and conducted the literature search along with the initial title/abstract review. JMY contributed her expertise in systematic review methodology to the design and conduct of this systematic review. LWL and JLB conducted the full text review, study selection, data extraction, and statistical analyses. LWL wrote the first draft of the manuscript. All authors contributed to critical review and approval of the final manuscript.

**Ethics Statement.** The authors adhered to the Preferred Reporting Items for Systemic Reviews and Meta-Analyses (PRISMA) guidelines. Ethical committee approval was not required for this study.

**PROSPERO registration number** CRD42019130459

**Data Availability Statement:** All data relevant to the study are included in the article or uploaded as supplementary information.

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## INTRODUCTION:

As thyroid function is frequently screened during the early gestational period, identification of thyroid immunity, defined by the presence of thyroid peroxidase antibodies (TPOAb) and/or thyroglobulin antibodies (TgAb), is relatively common in pregnant women. The prevalence of TPOAb positivity in the first trimester ranges from 5 to 15% in unselected pregnant women[1 2]. The prevalence of thyroid autoimmunity is even higher in women with a history of recurrent pregnancy loss, reported as 23 to 31%[3 4].

The association between thyroid autoimmunity and poor pregnancy outcomes, specifically miscarriage and preterm birth, has been well described[5 6]. However, the mechanism linking thyroid autoimmunity with increased risk of miscarriage and preterm birth in women without overt hypothyroidism is unclear. Two competing mechanistic theories propose that 1) thyroid autoantibodies are a marker of a hostile immune environment, or 2) thyroid autoantibodies act as a surrogate biomarker of impaired thyroid hormone reserves, despite normal thyroid function tests[7]. If the latter theory is true, one could postulate that supplementation with levothyroxine may improve thyroid hormone levels and prevent negative pregnancy outcomes.

Whether or not levothyroxine therapy, either prior to or during pregnancy, improves pregnancy outcomes in women with thyroid autoimmunity is controversial[8-14]. The 2017 American Thyroid Association (ATA) pregnancy guidelines strongly recommends levothyroxine for women with thyroid autoimmunity and a thyroid stimulated hormone (TSH) above the pregnancy-specific

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reference range. These guidelines recommend a consideration of levothyroxine in thyroid autoimmune women with TSH above 2.5 mIU/L and among euthyroid women with positive TPOAb and a history of pregnancy loss[15]. The evidence evaluating levothyroxine treatment in pregnant women with thyroid autoantibodies is conflicting as evidenced by three recent systematic reviews and meta-analyses[13 16 17]. The discrepant results may be attributed to the inclusion of small observational studies and the lack of inclusion of a recently published large multi-center trial[8]. Additionally, the limitations of observational studies must be considered. Observational studies identify associations, but do not provide the highest level of evidence for determining causation or therapeutic efficacy. Randomised controlled trials are considered the gold standard to evaluate therapeutic efficacy[18]. Therefore, we performed a systematic review and meta-analysis to evaluate all randomised controlled trials of levothyroxine therapy in women with thyroid autoimmunity who were pregnant or planning pregnancy.

## METHODS

### Protocol and Registration

A systematic review and meta-analysis was performed as outlined in the registered protocol (PROSPERO CRD42019130459), and is reported in accordance with the Preferred Reporting Items for Reviews and Meta-Analyses (PRISMA).

### Data Sources and Search Strategy

The following databases were searched in duplicate on April 17, 2019 (and updated on November 5, 2020) from 1980: Ovid MEDLINE, EMBASE, CINAHL, Cochrane Database of

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Systematic Reviews, and Cochrane Central Register of Controlled Trials. Clinicaltrials.gov was searched for active trials. The search strategy was developed in consultation with a medical librarian and used a combination of keywords related to hypothyroidism or thyroid autoimmunity and pregnancy (Supplemental S1). The search was conducted using an adapted version of the Cochrane search filter for randomised controlled trials where possible and was limited to human studies[19].

**Study Selection**

Studies that met all the following criteria were included in this review: (1) pregnant women or women contemplating pregnancy (including patients seeking assisted reproductive technologies), with positive TPOAb or TgAb, euthyroid or with subclinical hypothyroidism (by any prespecified study definition); (2) randomised to levothyroxine versus control (placebo or no treatment); (3) reported pregnancy outcomes; (4) reported in either English or French; (5) published from 1980 to November 5, 2020. The publication year 1980 was specifically chosen to coincide with the availability of more sensitive TSH assays[20 21].

Study selection was completed in two separate stages by two independent reviewers (PL and LED or LWL and JLB). First, duplicates were removed, and titles and abstracts were screened for eligibility (by PL and LED). Second, full citations deemed eligible at the title and abstract stage were retrieved and reviewed for eligibility in a full-text format (by LWL and JLB). Inclusion criteria was applied to select eligible articles and reasons for exclusion were recorded. Agreement was recorded at each stage and reported as a kappa statistic[22]. All disagreements between reviewers were solved by consensus, and, if needed, by discussion with a third

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independent reviewer (LED). Prior to each selection step, a pilot calibration was conducted by the respective reviewers to review inclusion criteria. The reference list of included articles was searched for the addition of further relevant studies.

**Data Collection Process**

Relevant study information was extracted by two independent reviewers (LWL and JLB) using standardized data extraction forms in Microsoft Excel (Version 16.16.18, Microsoft Corporation, Redmon, WA, USA). Extracted data elements included: study design, country, participant inclusion and exclusion criteria, TPOAb and TgAb results, definition of TPOAb and TgAb positivity, TPOAb and TgAb assays used, thyroid stimulating hormone (TSH) and thyroxine (T4) assays used and normal ranges, baseline serum concentrations of TSH and T4, iodine status of the country, levothyroxine dosing, achieved TSH and T4 after treatment, gestational timing of levothyroxine initiation, duration of treatment, details of the intervention, quality of the methods, inclusion of patients with subclinical hypothyroidism and definition of subclinical hypothyroidism, patient baseline characteristics, number of patients included in each group, pregnancy outcomes and their definitions.

**Quality Assessment**

Cochrane Risk of Bias Tool for Quality Assessment of Randomised Controlled Trials[23] was used to assess study quality and evaluate the risk of bias by two independent reviewers (LL and JB). This tool assesses seven methodologic domains and these components enable classification of randomised trials as “Low Risk of Bias”, “High Risk of Bias” or “Unclear Risk of Bias”.

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## Study Outcomes

The following pregnancy outcomes were prespecified and included: pregnancy by trial definition, miscarriage, preterm delivery, live-birth, birth weight, large for gestational age, small for gestational age, gestational age at delivery, multiple pregnancy, stillbirth, neonatal intensive care unit admission, preeclampsia, and gestational diabetes. GRADE (Grading of Recommendations, Assessment, Development and Evaluations) was utilized for grading of quality of evidence for the main outcomes[24].

## Data Synthesis and Analysis

Where appropriate, meta-analyses were conducted to evaluate the relative risk (RR) and 95% confidence intervals (95% CI) of pre-specified pregnancy outcomes including pregnancy achieved, miscarriage, preterm delivery and live-birth in participants randomised to levothyroxine therapy compared with placebo or no treatment. Random-effects models were used. Statistical heterogeneity was quantified using $I^2$ statistic, where $I^2$ >50% represents moderate and $I^2$ >75% represents substantial heterogeneity across studies[25]. The following prespecified sensitivity analyses were conducted to explore heterogeneity: 1) TSH levels at baseline, 2) time of levothyroxine initiation (preconception or during pregnancy), 3) use and type of assisted reproduction technologies, 4) maternal characteristics, and 5) study quality. Publication bias was assessed using a funnel plot as the graphical aid. Statistical analysis was performed using STATA version 14 (StataCorp, College Station, TX, USA) and Review Manager 5.3 (Version 5.3.5, The Cochrane Collaboration, Copenhagen, Denmark).

## Patient and Public Involvement

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No patient involved.

## RESULTS

The search results are summarised in Figure 1. Among the 3,023 citations reviewed for title and abstract, 79 citations were identified for full text review. Six randomised controlled trials (total of 2,263 women) met all eligibility criteria and were included in the qualitative and quantitative analyses[8-12 14]. There was a substantial level of agreement for both the title and abstract review ($\kappa$=0.64) and for the full text review ($\kappa$=0.70).

### Trial Characteristics

Trial characteristics are summarised in Table 1. Three trials initiated levothyroxine therapy prior to conception[8 11 14], whereas three trials initiated therapy in the first trimester[9 10 12]. Two trials compared levothyroxine to placebo[8 10], and the other four trials compared with no treatment[9 11 12 14]. Reproductive technologies were used in three trials[8 11 14], with one trial not specifying details with this regard. All trials included women with positive TPOAb with only one trial including TPO or Tg antibodies[14]. The TSH normal reference ranges and definition of TPO positivity were prespecified in each trial and are summarised in Table 1.

### Patient Characteristics

Baseline participant characteristics are summarised in Table 1 and Supplemental S2. Mean maternal age ranged from 26 years[9] to 32 years[8]. Mean reported body mass index ranged from 22.7 kg/m$^2$ [14] to 26.5 kg/m$^2$ [8]. Baseline TSH values varied according to assay specific reference ranges. Baseline free T4 values were comparable in the three trials that reported this information[8]

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[11 12] (Supplemental S2). There was insufficient information to assess the outcome of childhood neurodevelopment.

**Risk of Bias Assessment**

Risk of bias assessment is presented in Figure 2. Only one trial[8] was considered "low risk" of bias in all seven domains. Four trials[8 10-12] were assessed at "low risk" for selection bias. Two trials[9 14] were "unclear risk of bias" in allocation concealment. In four trials comparing levothyroxine to no treatment[9 10 12 14], the assessors were not blinded to patient randomization and thus these trials were considered to be "high risk" for performance bias. Published trial protocols were located for only two trials[8 14]. Thus, only these two trials were assessed as "low risk" for reporting bias.

**Pregnancy Outcomes**

There were sufficient data for meta-analyses of four pregnancy outcomes, as summarised in Figure 3. These outcomes include pregnancy achieved, miscarriage, preterm delivery and live births.

Pregnancy achieved was reported in three trials[8 11 14] (n=1,626) that initiated levothyroxine pre-conception. There was no significant difference in pregnancy achieved in thyroid autoimmune women treated levothyroxine compared with control (RR 1.03 [95% CI 0.93, 1.13]; $I^2$=0%). GRADE was rated as high quality for this outcome (Table 2).

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| | | | | | | | | Mean GA of LT4 initiation (weeks) | | Baseline TSH (mIU/L) (median, *mean) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Study Author (year)/ Country** | **N** | **Intervention levothyroxine** | **Control** | **Use of assisted reproduction technologies (% of participants)** | **Inclusion of subclinical hypothyroidism** | **TSH (Normal Range; mIU/L)** | **Defintion of TPOAb positivity (IU/mL)** | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm |
| Dhillon-Smith *et al.* (2019)/ United Kingdom | 952 | Initiated preconception | Placebo | Yes (45%) | No | 0.44-3.63 | > 99% concordance with UK NEQAS IIA | Preconception | Preconception | 2.1 | 2 |
| Nazarpour *et al.* (2017)/ Iran | 131 | Initiated 4-8 days following first prenatal visit | No treatment | Unclear | Yes | 0.1-2.5 | >50 | 11.4 ± 4.2 | 11.4 ± 4.2 | 3.7 | 3.2 |
| Negro *et al.* (2005)/ Italy | 72 | Initiated 1 month before assisted reproduction technologies | No treatment | Yes (100%) | Yes | 0.27-3.2 | >100 | Preconception | Preconception | 1.9* | 1.7* |
| Negro *et al.* (2006)/ Italy | 115 | Initiated at first endocrinological visit (2-7d after first obstetrics visit) | Placebo | No | No | 0.27-4.2 | >100 | 10.4 ± 3.1 | 10.3 ± 3.1 | 1.6* | 1.7* |
| Negro *et al.* (2016)/ Italy | 393 | Initiated in first trimester prior to 12 weeks | No treatment | No | No | 0.5-2.5 | >16 | 7.1 ± 1.2 | 7.0 ± 1.2 | 1.4* | 1.4* |
| Wang *et al.* (2017)/ China | 600 | Initiated 2-4 weeks prior to controlled ovarian hyperstimulation | No treatment | Yes (100%) | No | 0.45-4.78 | 60 | Preconception | Preconception | 2.9 | 2.1 |

**Table 1.** Characteristics of included randomised controlled trials. N, sample size; TSH, thyroid stimulating hormone; TPOAb, thyroid peroxidase antibody; LT4, levothyroxine; GA, gestational age; FT4, free thyroxine

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Miscarriage was reported in all six trials (n=1,427) and was defined as pregnancy loss at less than 20 weeks[9], less than 24 weeks[8], less than 28wks (early miscarriage defined as first 12 weeks, late miscarriage defined as between 13-28 weeks)[14], or not clearly defined[10-12]. There was no significant difference in the relative risk (RR) of miscarriage in TPOAb-positive pregnant women treated with levothyroxine compared with control (RR 0.93 [95% CI 0.76, 1.14]; $I^2$=0%). Inclusion of trials[8 11 14] that initiated levothyroxine only preconception revealed no significant difference compared with control (RR 0.91 [95% CI 0.72, 1.15]; $I^2$=0%). GRADE was rated as high quality for this outcome (Table 2).

Preterm delivery was reported in five trials (n= 1,354)[8-10 12 14]. Preterm delivery was defined as live birth prior to 34 weeks[8] or 37 weeks[9 10 12 14]. There was no significant difference in the relative risk of preterm delivery in thyroid autoimmune pregnant women treated with levothyroxine compared with control (RR 0.66 [95% CI 0.39, 1.10]; $I^2$=53%). GRADE was rated at moderate quality for this outcome. Downgrade considerations include considerable inconsistency, wide confidence intervals contributing to imprecision and evidence of publication bias.

Live births were reported in two trials (n= 1,540)[8 14]. Live birth was defined as delivery of a living fetus beyond 27 weeks[14] or beyond 34 weeks[8]. There was no significant difference in live births in thyroid autoimmune women treated with levothyroxine compared with control (RR 1.01 [95% CI 0.89, 1.16]; $I^2$ = 0%). GRADE was rated at high quality for this outcome (Table 2).

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| Outcome | Pregnancy achieved | Miscarriage | Preterm Delivery | Live births |
|---|---|---|---|---|
| **Risk of bias** | strong | strong | moderate | strong |
| **Imprecision** | strong | strong | moderate | strong |
| **Inconsistency** | strong | strong | weak | strong |
| **Indirectness** | strong | strong | strong | strong |
| **Publication bias** | strong | strong | weak | strong |
| **Overall GRADE** | high quality | high quality | moderate quality | High quality |

Table 2. Grading of Recommendations, Assessment, Development and Evaluations (GRADE) scoring of the quality of evidence for the main outcomes

Only one trial[10] reported on pre-eclampsia and found no difference in this outcome among thyroid autoimmune women treated with levothyroxine compared with control. None of the studies reported the outcome of gestational diabetes.

**Neonatal Outcomes:**

Neonatal outcomes were only reported in two trials (n = 1,083)[8 9]. Meta-analysis for gestational age at time of delivery (measured in days) and birth weight (measured in grams) are summarised in Supplemental S3. The weighted mean difference for gestational age at time of delivery in women treated with levothyroxine compared with placebo or no treatment was 2.35 days (95% CI -4.51, 9.20; $I^2$ = 90%). Weighted mean difference for birth weight in women treated with levothyroxine compared with placebo or no treatment was -24.05g (95% CI -96.91, 48.80; $I^2$ = 0%).

Only one trial[9] reported neonatal intensive care unit admission. They found fewer admissions among thyroid autoimmune women treated with levothyroxine (2 of 56 compared with control

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12 of 58, $p < 0.05$). No studies reported large for gestational age, small for gestational age or multiple pregnancy.

**Sensitivity Analyses**

A sensitivity analysis was performed to explore the heterogeneity of our meta-analysis for preterm delivery. Results are summarised in Supplemental S4.

TSH levels at baseline were subdivided into TSH ≤ 2.5 mIU/mL[8 10 12] and TSH > 2.5 mIU/mL[9 14]. The relative risk of preterm delivery in pregnant women with baseline TSH ≤ 2.5 mIU/mL was 0.64 (95% CI 0.36, 1.14; $I^2$=32%) compared with a relative risk of 0.62 (95% CI 0.17, 2.33; $I^2$=80%) in pregnant women with baseline TSH > 2.5 mIU/mL.

Timing of levothyroxine initiation was stratified into preconception[8 14] and in pregnancy[9 10 12]. The relative risk of preterm delivery in women initiated on levothyroxine preconception was 1.10 (95% CI 0.69, 1.75; $I^2$=0%) compared a relative risk of 0.46 (95% CI 0.28, 0.75; $I^2$=2%) in women initiated on levothyroxine in the first trimester.

The relative risk of preterm delivery in women using assisted reproductive technologies (ART)[8 14] was 1.10 (95% CI 0.69, 1.75, $I^2$ = 0%) compared with a relative risk of 0.51 (95% CI 0.26, 1.02; $I^2$ = 27%) in women that did not use ART[10 12].

Sensitivity analyses was also performed to explore the impact of fixed[8 11] and adjusted levothyroxine[9 10 12 14] doses on miscarriage rates (Supplemental S5). The relative risk of fixed

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levothyroxine dose was 0.89 (95% CI 0.66, 1.20; $I^2$=12%) compared with adjusted levothyroxine dose of 0.95 (95% CI 0.68, 1.33; $I^2$ = 0%).

Study quality was defined as high if there was low risk of bias across all seven domains of the Cochrane Collaboration's tool. We were unable to perform a sensitivity analysis exploring the impact of risk of bias on preterm delivery, as only one trial met the definition for high quality per the Cochrane Collaboration tool[8]. A funnel plot for preterm delivery is shown in Supplemental S6. On visual inspection, there was asymmetry indicating potential publication bias towards smaller studies reporting positive findings.

## DISCUSSION

This systematic review and meta-analysis of randomised controlled trials of levothyroxine therapy in women with thyroid autoimmunity without overt thyroid dysfunction demonstrates no evidence of benefit of levothyroxine therapy across prespecified pregnancy outcomes. Specifically, treatment with levothyroxine did not improve pregnancies achieved, miscarriage preterm delivery or live-birth across a total of 2,263 women, inclusive of those with a history of pregnancy loss and use of reproductive technologies. Furthermore, the availability of larger and better-quality trials since the publications of the American Thyroid Association 2017 pregnancy recommendations signals the need to update these guidelines as they pertain to the treatment of thyroid autoimmune women.

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Small randomised clinical trials initially demonstrated that levothyroxine therapy reduced miscarriage and/or pre-term delivery in TPOAb-positive women compared with placebo or no treatment intervention[9-11]. However, the unclear definition of miscarriage and pre-term delivery among these early trials means that selective reporting of outcomes could have biased their reported findings. Subsequent large, multi-center randomised controlled trials with clear prespecified outcome definitions have shown no impact of levothyroxine therapy on maternal or fetal outcomes compared with no levothyroxine treatment in pregnant women with subclinical hypothyroidism, isolated hypothyroxinemia or positive TPOAb[8 14 26 27]. These intervention trials involved pregnant women or women seeking pregnancy, including individuals with a history of pregnancy loss.

A systematic review and meta-analysis of a large study population (n = 47,045) of individual participant and population data demonstrated increased risk of preterm birth in TPOAb-positive women[6]. In light of these findings, Cappola and Casey[28] highlighted the need to address whether levothyroxine treatment decreases the risk of preterm birth among women with TPOAb. Three other systematic reviews and meta-analyses examined the effect of levothyroxine supplementation in pregnant women with thyroid autoimmunity[17 29 30]. Unlike our review which only included randomised controlled trials, two of the other reviews[24,25] included observational studies. The findings of our meta-analysis were consistent with the review by Sun *et al*.[17], which showed no benefit of levothyroxine supplementation on their pregnancy outcomes. In contrast, Rao *et al*.[29 30] demonstrated a beneficial effect of levothyroxine treatment on reducing pregnancy loss and preterm birth among pregnant women with subclinical hypothyroidism and/or thyroid autoimmunity. Several notable distinctions likely account for these discrepant results. First, the

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Dhillon-Smith *et al.*[8] publication was not included in these reviews. Furthermore, both reviews by Rao *et al.*[29 30] included a trial that strongly influenced their results but did not report outcome data in sufficient detail to be included in our systematic review[31]. Our attempts to clarify the outcome data of this trial[31] with the authors were left unresponded.

The findings of our study are consistent with the results of a recent systematic review and meta-analysis of randomised controlled trials of women with positive TPO antibodies [32]. These authors also conclude that levothyroxine supplementation is not associated with improvement in pregnancy outcomes. This further emphasizes the need to reconsider the current 2017 ATA guidelines on use of levothyroxine in euthyroid women with thyroid autoimmunity.

Most miscarriages occur early in pregnancy, prior to the first obstetrics appointment, thus initiation of levothyroxine prior to conception best addresses its impact on miscarriage rates. The pathophysiology underlying the association between thyroid autoimmunity in euthyroid pregnant women and miscarriage is still not completely understood. One hypothesis postulates that thyroid autoimmunity reflects inadequate thyroid reserves to accommodate for increased physiologic demands of pregnancy[33]. It has been postulated that delayed initiation of levothyroxine in pregnancy may account for its failure to reduce miscarriage. Our meta-analysis found that initiation of levothyroxine preconception or in the first trimester did not impact miscarriage outcomes. This suggests that inadequate thyroid reserve in women with thyroid autoimmunity does not account for the increased risk for miscarriage. A highly anticipated trial T4-LIFE that has not yet been published may further address the role of levothyroxine administration in women with TPOAb and recurrent miscarriage[34].

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Sensitivity analyses based on timing of levothyroxine initiation demonstrated a reduction in preterm delivery when levothyroxine was initiated in the first trimester rather than preconception. However, this finding is likely attributed to the quality of the included studies since many of the trials that initiated levothyroxine in the first trimester were at high risk of bias and the funnel plot suggests publication bias. Furthermore, it appears biologically implausible that levothyroxine would be beneficial for thyroid autoimmune women when initiated during pregnancy but not when initiated preconception.

A limitation of this study is that only one of the six included trials met all criteria for low risk of bias. Meta-analyses of live births and neonatal outcomes such as gestational age at delivery and birth weight were limited as only two trials reported these outcomes[8 9]. Furthermore, the sensitivity analysis is limited by the small number of studies included. Lastly, the funnel plot suggests publication bias towards smaller studies reporting positive findings. Interpretation of the funnel plot is limited by the small number of studies included in this meta-analysis[35].

This study has many notable strengths. First, this study provides an updated summary of randomised controlled trials not included in previous systematic reviews and meta-analyses or available at the time of the American Thyroid Association 2017 pregnancy guideline recommendations development. The systematic review and meta-analysis provides moderate (GRADE) evidence for the preterm delivery and high (GRADE) evidence for the outcomes of pregnancy achieved, miscarriage and live birth to guide clinical decisions. Our systematic review and meta-analysis benefits from the use of rigorous standardized tools such as Cochrane Risk of

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Bias tool[23] a pre-specified analysis plan posted on Prospero, sensitivity analysis that address study quality and the assessment of publication bias.

## CONCLUSION

In summary, data from six randomised controlled trials fail to demonstrate evidence of benefit from levothyroxine treatment for pregnancy outcomes in thyroid autoimmune women with normal thyroid function and subclinical hypothyroidism. In the absence of overt thyroid dysfunction, current trial evidence does not support the levothyroxine treatment of thyroid autoimmunity preconception or during pregnancy. Given the lack of evidence of benefit of levothyroxine therapy, it is time to re-examine recommendations for screening TPOAb and treatment of women with thyroid autoimmunity preconception or in pregnancy.

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**Figure and Table Legend:**

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Flow Diagram

Figure 2. Risk of bias assessment and trial quality assessment.

Figure 3. Results of meta-analysis of effects of levothyroxine treatment on pregnancy and obstetrical outcomes. (A) pregnancy achieved, (B) miscarriage (C) preterm delivery (D) live birth

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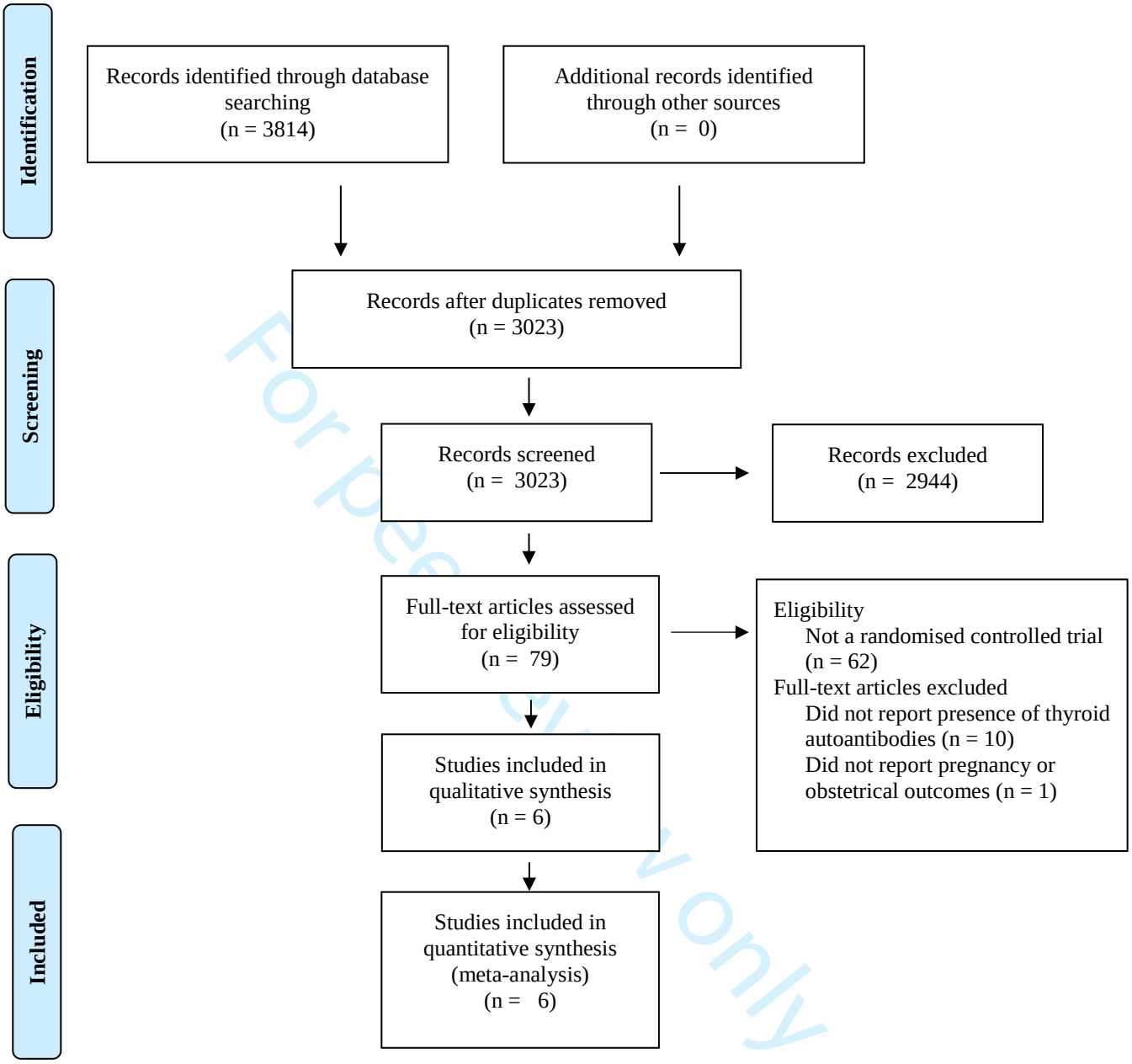
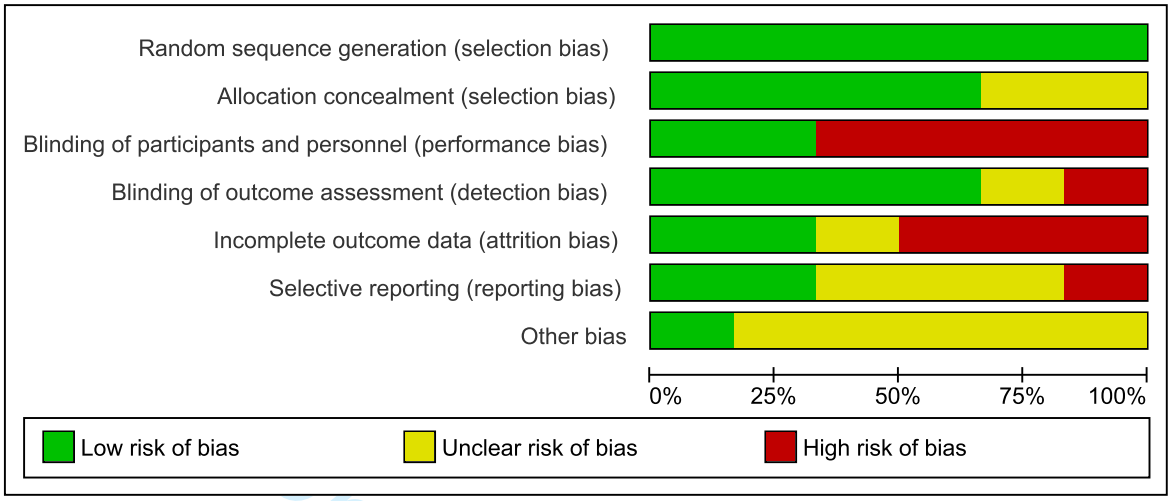



**Figure 1.** Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Flow Diagram

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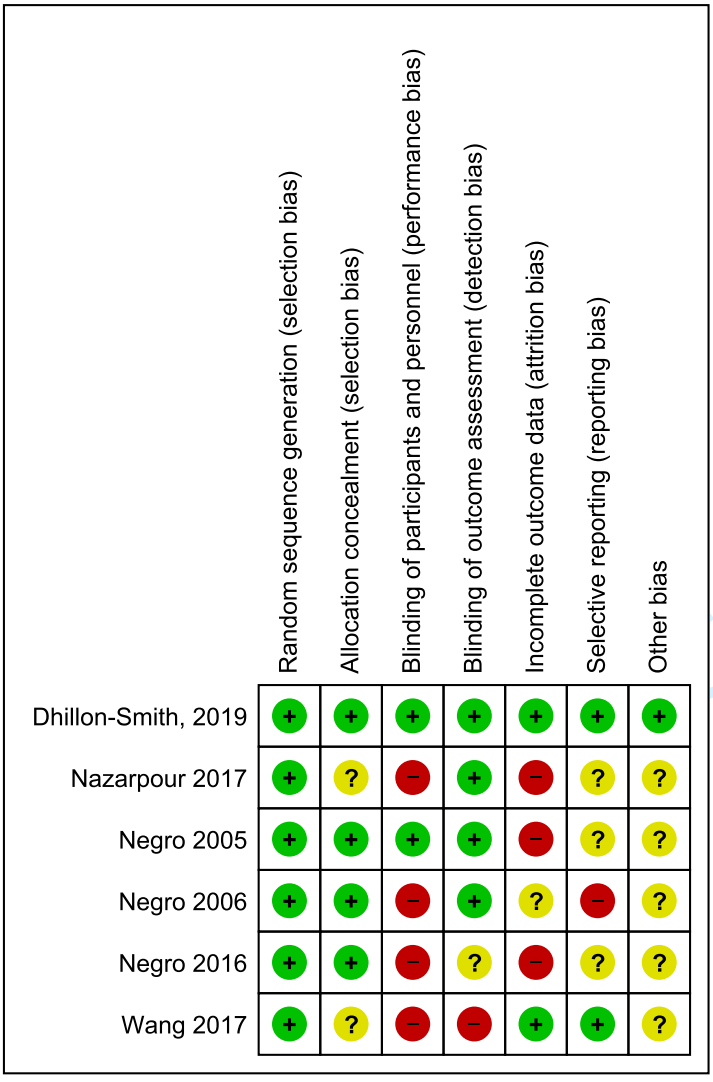
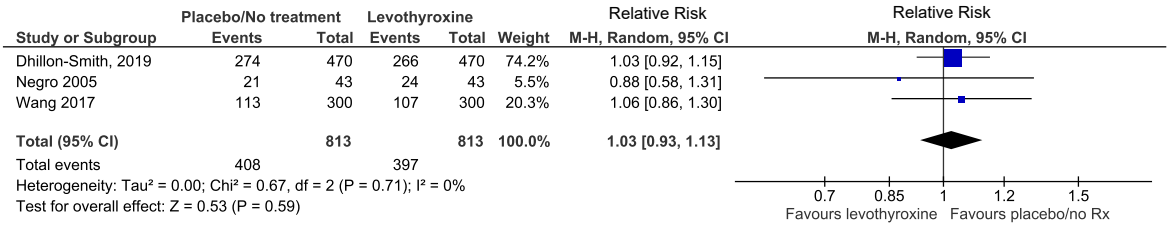
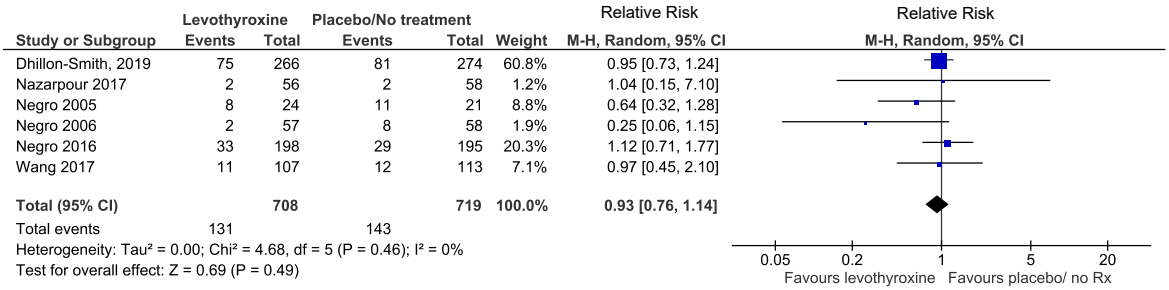




Figure 2. Risk of bias assessment and trial quality assessment.

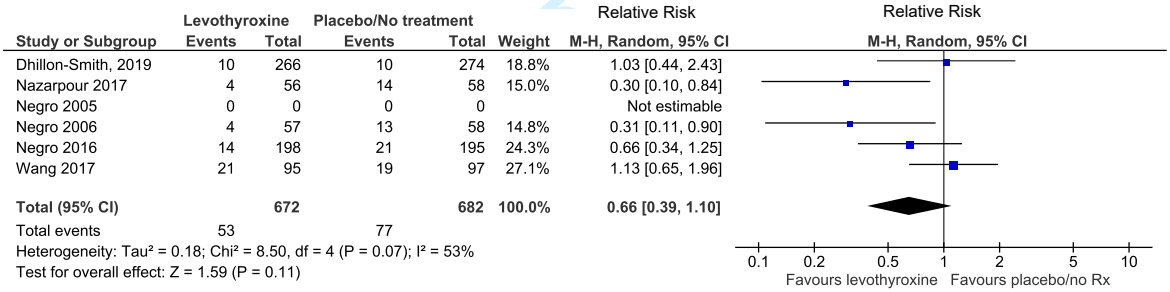
A. Pregnancy achieved




B. Miscarriage




C. Preterm delivery




D. Live births

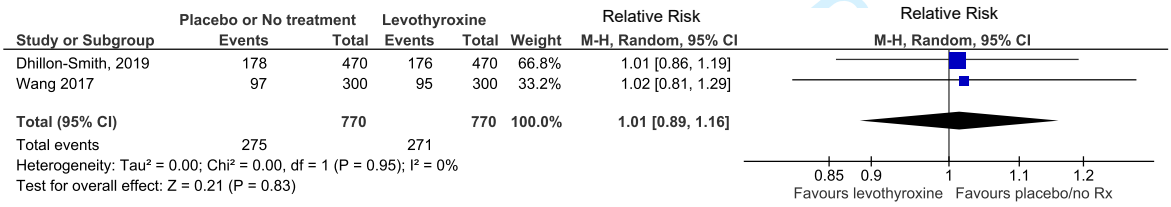



**Figure 3.** Meta-analysis of influence of levothyroxine on pregnancy outcomes. (A) pregnancy achieved, (B) miscarriage (C) preterm delivery (D) live birth

**Supplemental Information**

Supplemental Figure 1(S1): Full Search Strategy – MEDLINE (Ovid)

1. Hypothyroidism/
2. "hypothyroid*".ab,ti,kf.
3. "hypothyroxinemia".ab,ti,kf.
4. (("thyroid-stimulating hormone*" or "thyroid stimulating hormone*" or TSH*) adj2 deficien*).ab,ti,kf.
5. (thyroid* adj3 (autoimmun* or "auto-immun*" or antibod*)).ab,ti,kf.
6. Thyroxine/
7. "levothyro*".ab,ti,kf.
8. "thyrox*".ab,ti,kf.
9. synthroid.ab,ti,kf.
10. lt4.ab,ti,kf.
11. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10
12. exp Pregnancy/
13. "pregnan*".ab,ti,kf.
14. "gestat*".ab,ti,kf.
15. Pregnant Women/
16. "maternal*".ab,ti,kf.
17. "prenatal*".ab,ti,kf.
18. 12 or 13 or 14 or 15 or 16 or 17
19. 11 and 18
20. randomized controlled trial.pt.
21. controlled clinical trial.pt.
22. random*.ab.
23. placebo.ab.
24. drug therapy.fs.
25. trial.ab.
26. groups.ab.
27. 20 or 21 or 22 or 23 or 24 or 25 or 26
28. exp animals/ not humans.sh.
29. 27 not 28
30. 19 and 29
31. limit 30 to yr="1980 -Current"

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Supplemental Table (S2). Baseline patient characteristics

| | Maternal Age (years) | | Body Mass Index (kg/m$^2$) | | Baseline FT4 (pmol/L) | | Duration of Treatment | LT4 dosing and titration | Funding source |
|---|---|---|---|---|---|---|---|---|---|
| **Study Author (year)** | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm | Intervention Arm (LT4) | Control Arm | | | |
| Dhillon-Smith *et al.* (2019) | 32.5 ± 4.9 | 32.7 ± 4.9 | 26.4± 5.6 | 26.5± 5.5 | 14.6±1.9 | 14.5±2 | From before conception until the end of pregnancy | LT4 was d/c if TFT were outside assay specific limits at 6-8 wks, 1-18wks and 28 wks | UK national institute for health research |
| Nazarpour *et al.* (2017) | 26.6 ± 5.8 | 27 ± 4.7 | 24.9± 5.1 | 24.6 ± 3.6 | N/A | N/A | From the first trimester until the end of pregnancy | If TSH < 1.0 mcIU/mL: 0.5mcg/kg/d; If TSH 1.0-2.0, 0.75mcg/kg/d; If TSH > 2.0 or TPOAb > 1500IU/mL then 1mcg/kg/day | Did not receive any specific grant from any funding agency |
| Negro *et al.* (2005) | 29.2 ± 4 | 30.1 ± 5 | N/A | N/A | 11.2 ± 1.8 | 11.7 ± 0.7 | 1 month before assisted reproductive technology until the end of pregnancy | 1mg/kg/day | Unclear |
| Negro *et al.* (2006) | 30 ± 5 | 30 ± 6 | N/A | N/A | 14.8 ± 4.2 | 14.6 ± 4.3 | From the first trimester until the end of pregnancy | If TSH < 1.0 mcIU/mL: 0.5mcg/kg/d; If TSH 1.0-2.0, 0.75mcg/kg/d; If TSH > 2.0 or TPOAb > 1500IU/mL then 1mcg/kg/day | Unclear |
| Negro *et al.* (2016) | 28.9 ± 5.2 | 29.9 ± 5.1 | N/A | N/A | N/A | N/A | From the first trimester | If TSH 0.5-1.5, LT4 0.5mcg/kg/d; TSH 1.5-2.5, LT4 | Unclear |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | until the end of pregnancy | 1mcg/kg/d. Titration: If TSH > 3.0 in 2nd or 3rd trimester, LT4 dose increased by 12.5mcg/d; If TSH <0.5 in 2nd or 3rd trimester, then decrease by 12.5mcg/d | |
| Wang *et al.* (2017) | 31.3 ± 3.9 | 31.7 ± 3.8 | 22.7 ± 3.3 | 22.8 ± 3.2 | N/A | N/A | 2-4 weks before controlled ovarian hyperstimula-tion until the end of pregnancy | STARTING Dose: TSH >/= 2.5mIU/L: starting dose is 50mcg/d; TSH <2.5mIU/L, starting dose is 25mcg/d. If body weight <50kg, then starting dosing decreased by 50%<br><br>Titration: TSH in first trimester 0.1-2.5mIU/L, 0.2-3.0mIU/L in second trimester, 0.3-3.0mIU/L in third trimester | National Key Technology R and D program and Chinese National 973 program (Both form the Ministry of Science and Technology of China) |

Table S2: N/A specifies that information was not available. FT4 denotes free thyroxine. LT4 denotes levothyroxine.

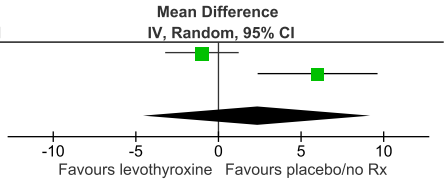
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## Supplemental Figure (S3):

### A. Weighted mean difference for gestational age at time of delivery (days)

| Study or Subgroup | Levothyroxine Mean | SD | Total | Placebo or No Rx Mean | SD | Total | Weight | Mean Difference IV, Random, 95% CI |
|---|---|---|---|---|---|---|---|---|
| Dhillon-Smith, 2019 | 272 | 17 | 476 | 273 | 18 | 476 | 52.2% | -1.00 [-3.22, 1.22] |
| Nazarpour 2017 | 275 | 9 | 65 | 269 | 12 | 66 | 47.8% | 6.00 [2.37, 9.63] |
| **Total (95% CI)** | | | **541** | | | **542** | **100.0%** | **2.35 [-4.51, 9.20]** |

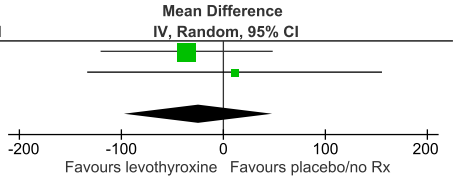
Heterogeneity: Tau² = 22.14; Chi² = 10.39, df = 1 (P = 0.001); I² = 90%
Test for overall effect: Z = 0.67 (P = 0.50)




### B. Weighted mean difference for birth weight (g)

| Study or Subgroup | Levothyroxine Mean | SD | Total | Placebo/No Rx Mean | SD | Total | Weight | Mean Difference IV, Random, 95% CI |
|---|---|---|---|---|---|---|---|---|
| Dhillon-Smith, 2019 | 3,226 | 660 | 476 | 3,262 | 668 | 476 | 74.6% | -36.00 [-120.36, 48.36] |
| Nazarpour 2017 | 3,139 | 288 | 65 | 3,128 | 524 | 66 | 25.4% | 11.00 [-133.51, 155.51] |
| **Total (95% CI)** | | | **541** | | | **542** | **100.0%** | **-24.05 [-96.91, 48.80]** |

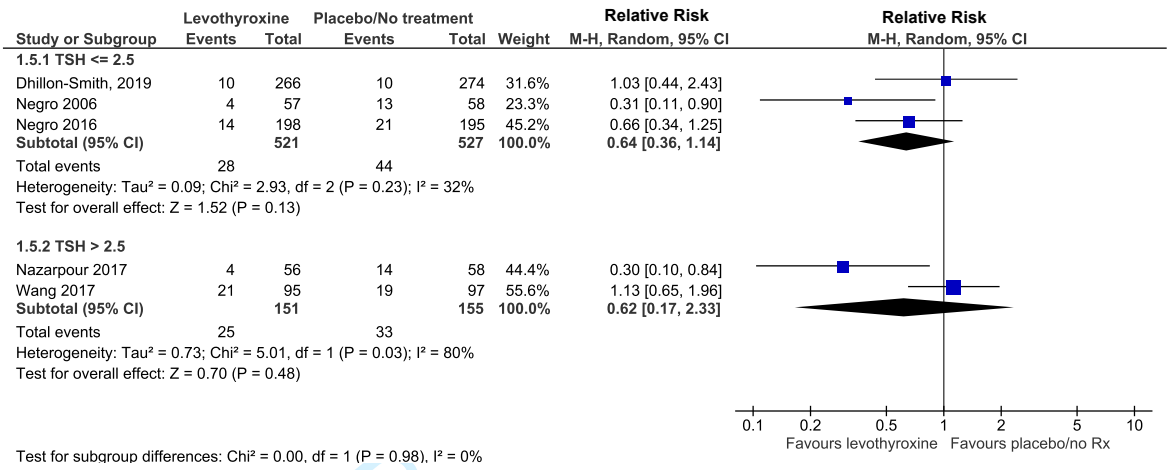
Heterogeneity: Tau² = 0.00; Chi² = 0.30, df = 1 (P = 0.58); I² = 0%
Test for overall effect: Z = 0.65 (P = 0.52)



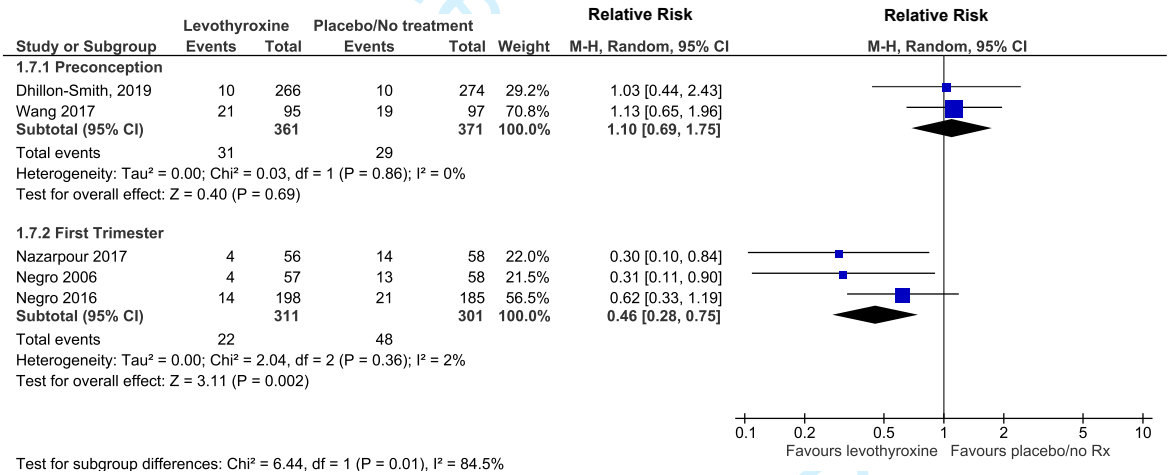

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Supplemental Figure (S4): Stratified meta-analysis for preterm delivery subgroups.

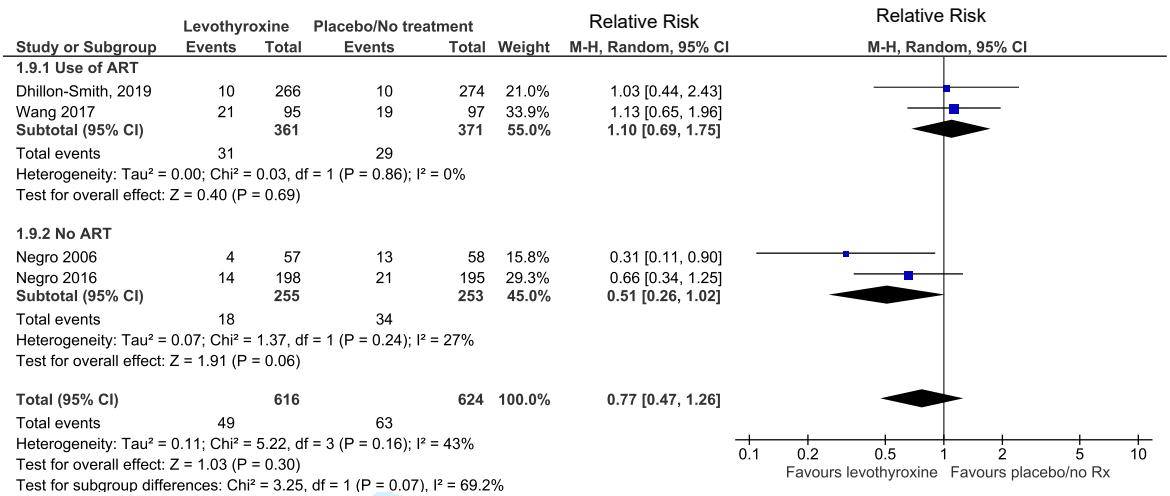
## A. Stratified by TSH levels at baseline



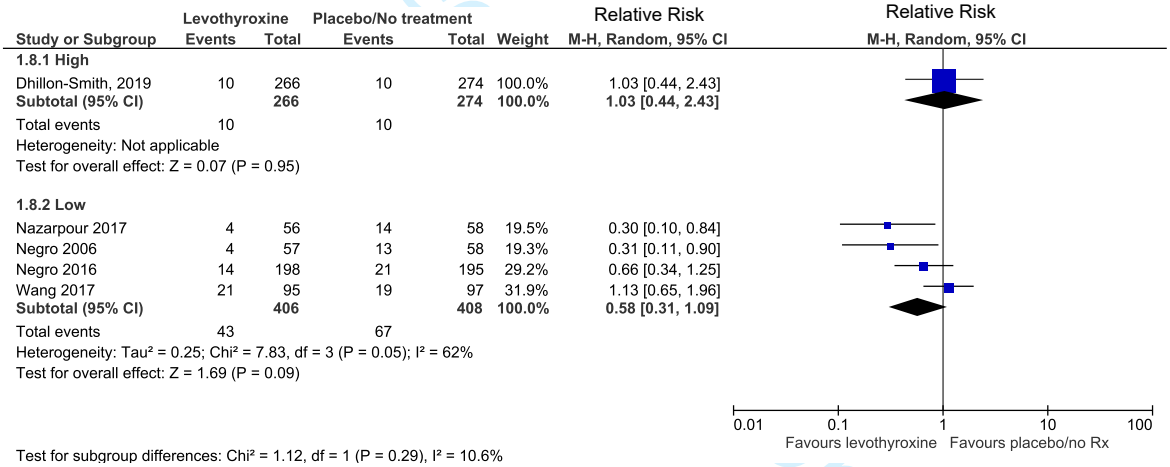

## B. Stratified by timing of levothyroxine initiation.



## C. Stratified by use of assisted reproductive technologies

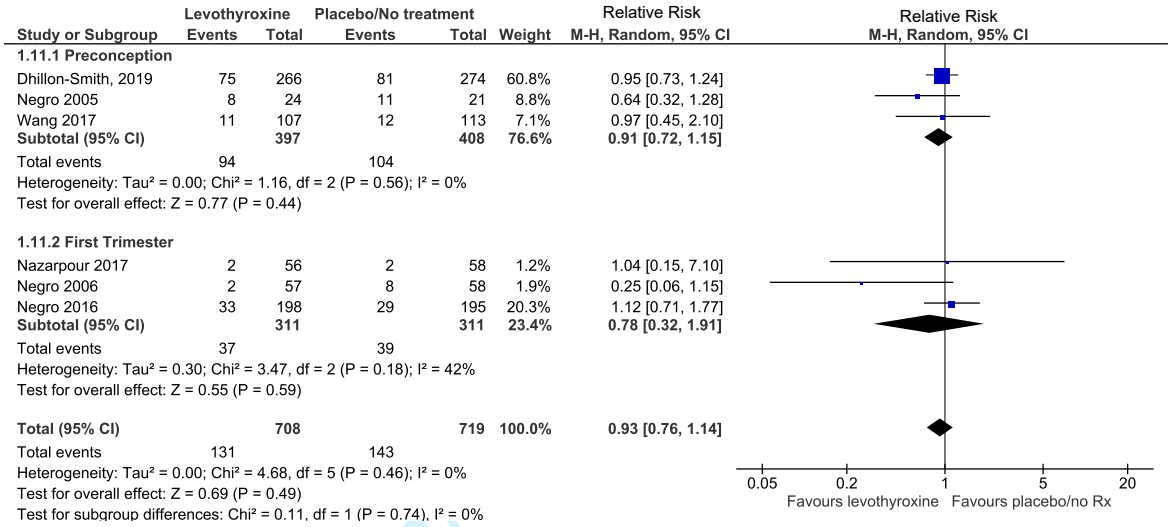



## D. Stratified by study quality

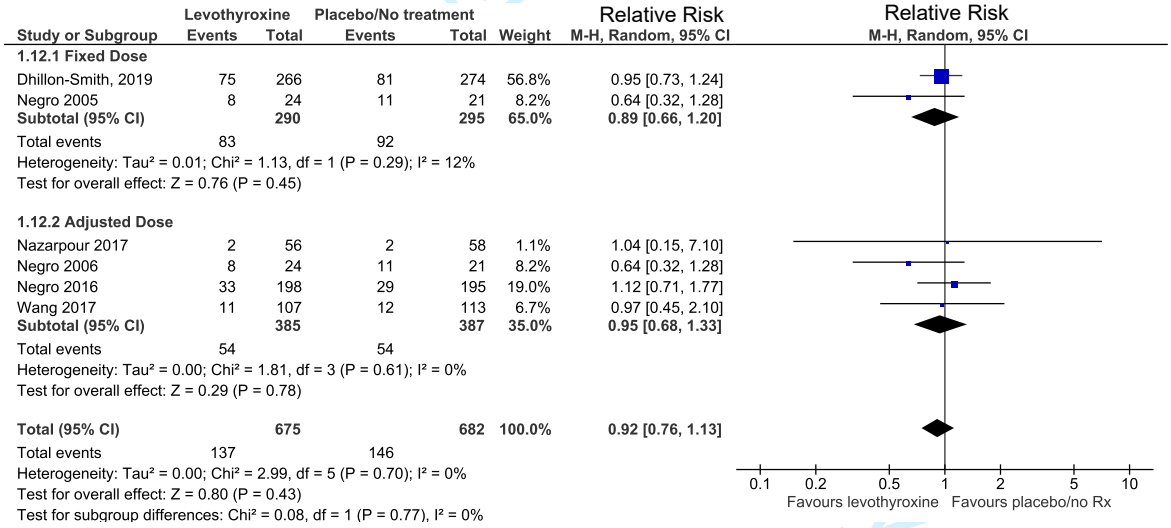


Supplemental Figure (S5): Stratified meta-analysis for miscarriage subgroups.

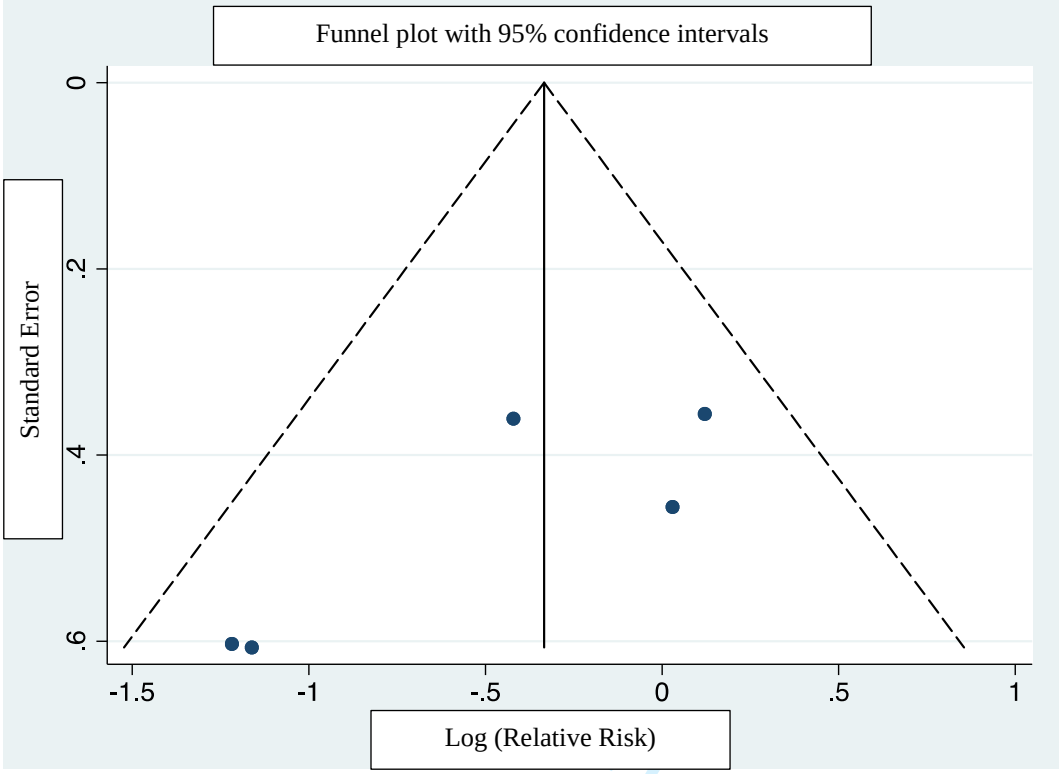
A. Stratified by timing of levothyroxine initiation.




B. Stratified by fixed levothyroxine versus adjusted levothyroxine dose.



Supplemental Figure (S6): Funnel plot for preterm delivery




view only

# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 4 | Describe the rationale for the review in the context of what is already known. | 3-4 |
| Objectives | 4-5 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 5 |
| Eligibility criteria | 5 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 5 |
| Information sources | 5 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 5 |
| Search | 5 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 5-6 |
| Study selection | 6 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 6 |
| Data collection process | 6-7 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 6 |
| Data items | 7 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 6 |
| Risk of bias in individual studies | 7 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 7 |
| Summary measures | 7-8 | State the principal summary measures (e.g., risk ratio, difference in means). | 7-8 |
| Synthesis of results | 8 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 8 |

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# PRISMA 2009 Checklist



| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 7 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 7 |
| Additional analyses | 8 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 8 |
| **RESULTS** | | | |
| Study selection | Fig 1 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | Figure 1 |
| Study characteristics | Table 1 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | Table 1 |
| Risk of bias within studies | Fig 2 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | Figure 2 |
| Results of individual studies | Fig 3 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | Figure 3 |
| Synthesis of results | Fig 3 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | Figure 3 |
| Risk of bias across studies | Fig 2 | Present results of any assessment of risk of bias across studies (see Item 15). | Figure 2 |
| Additional analysis | Suppl | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | Appendix |
| **DISCUSSION** | | | |
| Summary of evidence | 13 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 13 |
| Limitations | 15-16 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 15 |
| Conclusions | 16 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 16 |
| **FUNDING** | | | |
| Funding | 3 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 1 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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